

# Software Requirements Specification

SYD466 – 2026 Winter

NAA – Group 3

Name of Project: Chocolatetown Business Management System (CBMS)

Project Leader: Anderson Yu-Hong Cai

Last updated: 2026/04/12

## Team Members

| Student ID | Name                 | Email                                                      |
|------------|----------------------|------------------------------------------------------------|
|            | Anderson Yu-Hong Cai |                                                            |
|            | Jay Hsia             |                                                            |
|            | Masanori Nishimura   |                                                            |
| 173968231  | Yi-Lun Wu            | <a href="mailto:ywu296@myseneca.ca">ywu296@myseneca.ca</a> |

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# 1 Introduction/Overview - Document Information

## 1.1 Document Authors

1. Anderson Yu-Hong Cai, 109262246
2. Yi-Lun Wu, 173968231
3. Masanori Nishimura, 102621240
4. Jay Hsia, 112935242

## 1.2 Revision History

| Weeks | Sections of this document that were completed/updated this week:                                                                                                                    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week1 | Section 1, Section 2, Sections 3.1 and 3.2                                                                                                                                          |
| Week2 | Section 3.3 - Section 3.6, Section 4                                                                                                                                                |
| Week3 | Section 5.1 - Section 5.2                                                                                                                                                           |
| Week4 | Section 5.3.1 - Section 5.3.8:<br>Masanori - US-PS-01, US-PS-04    Anderson - US-CM-01, US-CO-06<br>Yi-Lun - US-CO-01, US-PM-01    Jay - US-SM-01, US-SM-04<br>Section 6 - version1 |
| Week5 | Section 5.3.1 - Section 5.3.8:<br>Masanori - US-PS-02, US-PS-05    Anderson - US-CM-02, US-CO-07<br>Yi-Lun - US-CO-02, US-PM-02   Jay - US-SM-05, US-SM-06<br>Section 6 - version2  |
| Week6 | Editing & Proofreading                                                                                                                                                              |
|       |                                                                                                                                                                                     |
|       |                                                                                                                                                                                     |

## 1.3 Document Conventions

Any **text in red** indicates template or notes.

Any **text in blue** is in-progress.

Any **text in green** was recently added.

Any **text highlighted in yellow** is an important point.

Any *text italicized* represents definitions.

## 1.4 Document Purpose

The purpose of this Software Requirements Specification (SRS) document, developed for the SYD466 course, is to clearly outline the functionalities, design, and constraints of the Chocolatetown Business Management System for the SYD466 project. This document serves as a

formal guide for all stakeholders, including the project team (Programmers, UX/UI Designers, Business Analysts, Testers), and the client, sponsor, and users (Seneca, course professor), ensuring a mutual understanding of the system's objectives and requirements.

Specifically, this SRS will:

- Define a detailed functional baseline for the Chocolatetown Business Management System to ensure all business needs, such as staff payments and catering invoices, are accurately captured.
- Establish the behavioral specifications of the system across various operational scenarios, providing a clear roadmap for the design of Domain and Entity layers.
- Specify technical, operational, and non-functional requirements to ensure the system remains secure, scalable, and mobile-accessible as required by the client.
- Serve as the primary technical blueprint throughout the SDLC, aligning the final implementation with both the course design standards (MVC) and Mike's business objectives.
- Document all project constraints, assumptions, and dependencies to mitigate risks related to financial compliance and CRA auditing.

This document is intended for use by all the stakeholders involved in the project's development and evaluation.

## 1.5 Intended Audience

This section identifies the intended audience for this document. It outlines the key stakeholders who will use or refer to the system requirements, along with their roles and purposes. Defining the intended audience helps ensure the document is clear, relevant, and useful for everyone involved in the project.

| Stakeholder Category             | Description                                                                                         | Purpose                                                                                            |
|----------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Project Sponsor / Business Owner | The owner or manager responsible for overseeing business operations and making strategic decisions. | To ensure the system supports business goals and operational needs.                                |
| Project Manager                  | The person responsible for planning, coordinating, and monitoring the project.                      | To track project progress and ensure requirements and timelines are met.                           |
| Development Team                 | Developers responsible for designing, implementing, and maintaining the system.                     | To understand system requirements and technical specifications for development.                    |
| QA Team                          | Quality Assurance specialists responsible for testing the system to ensure it meets                 | To understand system requirements and expected behavior in order to design and execute test cases. |

|  |                                       |  |
|--|---------------------------------------|--|
|  | requirements and functions correctly. |  |
|--|---------------------------------------|--|

## 2 Group Agreement

### TEAM AGREEMENT

**Team #:** Group 3

**Project Title:** Chocolatetown Business Management System (CBMS)

**Project Time Frame:** March 2026 - April 2026

**Team Members:** Anderson Yu-Hong Cai, Jay Hsia, Masanori Nishimura, Yi-Lun Wu

**Team Leadership:** Anderson Yu-Hong Cai

**Team Functions:** The team will collaborate on all sections of the SRS, while each member takes primary responsibility for a specific focus area.

(1) Anderson Yu-Hong Cai: Focus on customer and catering management, including catering bookings and customer information management. Responsible for coordinating project progress, integrating the final SRS document, and overseeing system integration requirements.

(2) Yi-Lun Wu: Focus on customer and catering operations, including catering orders and invoice-related requirements. Also responsible for documenting product and menu-related information.

(3) Masanori Nishimura: Focus on staff management and payroll-related requirements. Also responsible for documenting system accessibility requirements, including mobile and laptop access.

(4) Jay Hsia: Focus on supplier management and supplier transaction requirements. Also responsible for documenting system security and privacy requirements.

We will share information through: Microsoft Teams, WhatsApp group chat

**Team Meetings:** One mandatory meeting per week (schedule to be decided weekly in WhatsApp), plus daily asynchronous updates.

### **Team Problems:**

(1) If any issues arise, the team will first discuss the problem internally. If the issue cannot be resolved, the team leader will inform the instructor.

(2) To ensure the project can be delivered on time, the team will hold weekly meetings and provide daily asynchronous updates to track progress.

(3) To ensure project quality, the team will use weekly meetings to share opinions and align expectations.

**Team Commitment:** The team is committed to participating in all meetings. If any unexpected issues arise, they should be communicated to the team lead. All participants must complete their assigned parts.

**The undersigned members agree to work together on the project. They recognize that as a team and individually they are responsible for the quality of all deliverables.**

| Name                 | Date     |
|----------------------|----------|
| Anderson Yu-Hong Cai | 2026/3/5 |
| Yi-Lun Wu            | 2026/3/5 |
| Jay Hsia             | 2026/3/5 |
| Masanori Nishimura   | 2026/3/5 |

## **3 Project Overview**

### **3.1 Problem Statement**

|                      |                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem:</b>      | Informal and fragmented business operations, including manual staff payments, inconsistent supplier transaction tracking, and ad hoc catering invoice management. |
| <b>Affects:</b>      | The business owner, staffs, catering customers, suppliers, and financial institutions such as the bank.                                                           |
| <b>Impact:</b>       | Growing inefficiencies, lack of structure, and compliance risks related to financial tracking and record-keeping, as highlighted by a recent CRA audit.           |
| <b>Future State:</b> | A centralized, reliable, and scalable software solution that manages operations effectively, integrates with existing tools, and ensures strong data security.    |

## 3.2 Product Vision

|                         |                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>For</b>              | The owner of Chocolatetown.                                                                                                                                |
| <b>Who</b>              | Needs to manage growing business operations, payments, and invoices in an organized and auditable way.                                                     |
| <b>The Product Type</b> | A centralized Business Management System.                                                                                                                  |
| <b>That</b>             | Manages staff payments, supplier transactions, and catering invoices while integrating with his existing POS system and website.                           |
| <b>Unlike</b>           | The current manual, fragmented, and ad hoc record-keeping methods.                                                                                         |
| <b>Our product</b>      | Is accessible via mobile devices and laptops, ensures strong data privacy for client information, and is scalable for future catering and location growth. |

### Product Vision Statement

**For** Mike, the owner of Chocolatetown, **who** needs to centralize and secure his fragmented business operations, the **Chocolatetown Business Management System** is a web and mobile-accessible platform **that** streamlines staff payments, supplier transactions, and catering invoices to ensure strict financial compliance. **Unlike** the current manual and ad-hoc record-keeping methods that pose risks during CRA audits, **our product** provides a secure, scalable, and real-time management solution **because** it integrates all core business functions into a single, auditable system accessible via both laptops and mobile devices.

## 3.3 Stakeholders and Users

| Stakeholder Name/Identifier  | Category                                   | Description                                                                                                              | Purpose                                                                                                                                     |
|------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Business Owner</b>        | Primary Stakeholder / System Administrator | Owner of Chocolatetown who oversees all business operations including finances, staff management, and catering services. | Uses the system to manage staff payments, track supplier transactions, monitor catering invoices, and oversee overall business performance. |
| <b>Store Staff</b>           | Internal User                              | Employees working in the store who assist customers, process catering orders, and help record transactions.              | Use the system to record catering orders, update order status, and assist with customer service operations.                                 |
| <b>Payroll Administrator</b> | Internal User                              | Responsible for managing staff profiles, payments and compensation records. Although this function is                    | Use the system to manage staff profiles, track employee payment records, salaries, and                                                      |



|                                      |                        |                                                                                                                                              |                                                                                                                                                                               |
|--------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      |                        | currently handled by the owner, the system supports this role to allow future delegation as the business grows.                              | payment history in an organized way.                                                                                                                                          |
| <b>Supplier Coordinator</b>          | Internal User          | Responsible for managing supplier relationships and handling supplier-related transactions, including supplier invoices and payment records. | Use the system to record supplier invoices, track supplier transactions, and process payments to suppliers in an organized and auditable way.                                 |
| <b>Catering Customers</b>            | External Stakeholder   | Customers who request catering services for events such as birthdays, corporate gatherings, and private parties.                             | Provide event requirements and receive catering services, invoices, and confirmations through the business.                                                                   |
| <b>Suppliers</b>                     | External Stakeholder   | Companies that provide ingredients, packaging materials, and other supplies used by Chocolatetown.                                           | Provide goods and services to the business. Supplier invoices and payment transactions are recorded and tracked in the system for financial management and auditing purposes. |
| <b>Financial Institutions (Bank)</b> | External Stakeholder   | Banks or payment processors used for financial transactions and payment processing.                                                          | Support secure payment processing and financial transaction verification.                                                                                                     |
| <b>Government / CRA</b>              | Regulatory Stakeholder | The Canada Revenue Agency that audits and ensures tax compliance for businesses.                                                             | Require accurate financial records and transaction tracking for regulatory compliance and auditing purposes.                                                                  |

### 3.4 Functional Requirements

| ID                                                 | Title                     | Description                                                                                                                                                                                                              | Priority Level [0 – 100/ highest] |
|----------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Payroll &amp; Staff Management</b>              |                           |                                                                                                                                                                                                                          |                                   |
| FR001                                              | Staff Registration        | The system shall allow the Payroll Administrator to register new staff members in the system by entering their name, contact info, and role.                                                                             | 95                                |
| FR002                                              | Payment Record            | The system shall enable the Payroll Administrator to record staff payments by specifying the amount, date, and payment method.                                                                                           | 100                               |
| FR003                                              | Payment History           | The system should allow the Payroll Administrator to view a staff member's payment history for a selected period, including payment records processed and transferred through external payment systems where applicable. | 80                                |
| FR004                                              | Payroll Report            | The system should generate a monthly payroll summary report for review and record-keeping purposes based on payroll records and transaction data.                                                                        | 80                                |
| <b>User Management</b>                             |                           |                                                                                                                                                                                                                          |                                   |
| FR005                                              | User Registration         | The system shall allow authorized personnel to register a new user account by providing required information such as staff ID, name, role, and password.                                                                 | 100                               |
| FR006                                              | User Login                | The system shall allow user to log into the system by their staffid and password                                                                                                                                         | 100                               |
| FR007                                              | Role-based Access Control | The system shall restrict access to system features based on the user's assigned role.                                                                                                                                   | 85                                |
| FR008                                              | User Logout               | The system shall allow users to securely log out of the system.                                                                                                                                                          | 90                                |
| FR009                                              | User Account Update       | The system shall allow authorized personnel to update user account information.                                                                                                                                          | 85                                |
| FR010                                              | User Account Deactivation | The system shall allow authorized personnel to deactivate user accounts without deleting historical data.                                                                                                                | 90                                |
| <b>Catering Operations and Customer Management</b> |                           |                                                                                                                                                                                                                          |                                   |
| FR011                                              | Manage Customer Profiles  | The system shall allow Store Staff and Business Owner to create and update                                                                                                                                               | 90                                |

|       |                                         |                                                                                                                                                                                                                                                                                                 |     |
|-------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|       |                                         | catering customer profiles by entering customer contact information, including information received from catering inquiries submitted through the Existing Website.                                                                                                                             |     |
| FR012 | View Customer Report                    | The system shall allow Store Staff and Business Owner to view customer reports, including customer profiles, catering history, and contact information, to support business analysis and decision-making.                                                                                       | 85  |
| FR013 | Create Customer Records from Inquiries  | The system shall allow Store Staff to create customer profiles and catering records based on data received from catering inquiries.                                                                                                                                                             | 90  |
| FR014 | Manage Catering Order                   | The system shall allow Store Staff to create, update, and delete a catering order by selecting a customer profile, menu items, and event date and details. The system shall also allow Store Staff to create catering orders based on catering inquiries received through the Existing Website. | 100 |
| FR015 | View Catering Order                     | The system may allow Store Staff to view a calendar or list of upcoming catering events by filtering dates.                                                                                                                                                                                     | 80  |
| FR016 | Receive Catering Inquiries from Website | The system shall receive catering inquiries submitted through the Existing Website.                                                                                                                                                                                                             | 95  |
| FR017 | Update Catering Order Status            | The system shall allow Store Staff to update the status of a catering order (e.g., pending, preparing, ready, delivered) to track progress.                                                                                                                                                     | 85  |
| FR018 | Generate Catering Invoice               | The system shall allow the Store Staff to generate a detailed invoice for a catering order by automatically calculating the total cost of selected items.                                                                                                                                       | 100 |
| FR019 | Track Invoice Payment Status            | The system shall allow the Store Staff to update and track the payment status (e.g., unpaid, partially paid, fully paid) of catering invoices.                                                                                                                                                  | 100 |
| FR020 | Retrieve POS Transaction Data           | The system shall retrieve transaction data from the Existing POS System, including sales records and transaction amounts, to support financial reporting and audit requirements.                                                                                                                |     |
| FR021 | Manage Menu Items                       | The system shall allow the Business Owner to create, update, and delete product and                                                                                                                                                                                                             | 85  |

|                            |                              |                                                                                                                                                                                             |     |
|----------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|                            |                              | menu items by entering the item for details, descriptions, and pricing.                                                                                                                     |     |
| <b>Supplier Management</b> |                              |                                                                                                                                                                                             |     |
| FR022                      | Supplier Management          | The system shall allow the Supplier Coordinator to create, update, view, and delete supplier records, including supplier name, contact information, payment details, and supplied products. | 95  |
| FR023                      | Supplier Information         | The system should store supplier information including supplier name, contact information, payment details, and supplied products.                                                          | 80  |
| FR024                      | Record Supplier Transactions | The system shall allow Supplier Coordinator to record supplier transactions such as ingredient purchases or material supplies.                                                              | 100 |
| FR025                      | Supplier Invoice Record      | The system shall allow Supplier Coordinator to upload or record supplier invoices associated with each transaction.                                                                         | 85  |
| FR026                      | Record Supplier Payment      | The system shall allow the Supplier Coordinator to record payments made to suppliers, including payment amount, payment date, and payment method.                                           | 95  |

### 3.5 Non-functional Requirements

| ID     | Title                      | Description                                                                                                                                                                     |
|--------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR001 | Security / Privacy         | The system shall ensure data security by encrypting all customer contact information and catering order records to protect sensitive client information in case of device loss. |
| NFR002 | Compliance                 | The system shall support regulatory compliance by accurately retaining all catering invoice and transaction data to meet Canada Revenue Agency (CRA) auditing standards.        |
| NFR003 | Authorization              | The system shall restrict access to supplier management and financial records based on user roles (e.g., owner or authorized staff only).                                       |
| NFR004 | Audit Logging              | The system shall record key actions such as supplier creation, transaction updates, and payment records for auditing purposes.                                                  |
| NFR005 | Multi-device Accessibility | The system shall provide a responsive web interface that is fully accessible through mobile devices (iOS/Android) and laptops (Windows/macOS).                                  |

|        |              |                                                                                                                                                         |
|--------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR006 | Performance  | The system shall maintain high performance by generating and displaying catering invoices within 2 seconds of the user submitting the request.          |
| NFR007 | Scalability  | The system shall support increased numbers of users, catering orders, and suppliers as the business grows.                                              |
| NFR008 | Data Privacy | The system shall protect sensitive business information such as supplier details, payment records, and financial transactions from unauthorized access. |
| NFR009 | Data Backup  | The system shall maintain regular backups of supplier transaction data to prevent data loss.                                                            |

### 3.6 Interaction with External Systems



The Chocolatetown Business Management System is designed to integrate with key existing systems to support business operations.

The system retrieves transaction data from the Existing POS System to support financial record keeping and reporting.

In addition, the system receives catering inquiries from the Existing Website, which are used to create and manage customer profiles and catering service records.

These integrations ensure that business data is centralized, accurate, and efficiently managed within the system.

## 4 Project Scope

### 4.1 Purpose and Objective

#### 4.1.1 Purpose Statement

The purpose of this project is to develop the **Chocolatetown Business Management System (CBMS)** for the owner and staff of Chocolatetown. The system helps the business manage daily operations such as staff payments, supplier transactions, and catering invoices. It replaces the current manual and fragmented process with a centralized digital system.

#### 4.1.2 Objectives

The system aims to solve the problem of unorganized business operations and inconsistent financial records. By using a centralized system, the business owner can manage suppliers, staff payments, catering orders, and invoices more efficiently.

#### SMART Objectives

- To allow the Business Owner and Store Staff to manage catering orders and customer information in an organized way during daily operations.
- To record supplier transactions and supplier payments for better financial tracking.
- To store staff payment records so the Business Owner can review payroll information when needed.
- To generate catering invoices automatically when catering orders are created.
- To provide secure system access using user login and role-based access control.

### 4.2 In Scope Features

- User Login and Access Control – Users can log into the system and access features based on their roles.
- Staff and Payroll Management – The system records staff information and staff payment records.
- Customer Profile Management – The system stores catering customer information and contact details.
- Catering Order Management – Store Staff can create catering orders and update order status.
- Catering Invoice Generation – The system generates invoices and tracks invoice payment status.
- Supplier Management – The system stores supplier information and supplier contact details.

- Supplier Transaction Management – The system records supplier invoices and supplier payments.
- Menu and Product Management – The Business Owner can manage dessert menu items and pricing.
- Financial Record Tracking – The system stores financial records related to payroll, supplier payments, and catering invoices.
- Integration with Existing Systems – The system receives catering inquiries from the website and transaction data from the POS system.

### **4.3 Out of Scope Features**

- Customer Self-Service Portal – Catering customers cannot access the system directly.
- Online Payment Processing – The system will not process credit card or online payments.
- Inventory Management System – Ingredient inventory tracking is not included.
- Advanced Financial Analytics – Business intelligence or predictive analytics features are not included.
- Native Mobile Application – The system will be a web application, not a mobile app.

### **4.4 Constraints**

- The system must be completed within the planned project development timeline.
- The development team consists of four student developers.
- The system must work on modern web browsers and support both laptops and mobile devices.
- The system must follow data security and privacy requirements to protect financial records.
- Integration with the POS system and website may be limited to basic data exchange.

### **4.5 Assumptions**

- The Business Owner and Store Staff have basic computer skills and can use a web-based system.
- Supplier information, customer data, and financial records provided by the business are accurate.
- The business website and POS system will continue to operate normally.
- Users will access the system using stable internet connections and modern web browsers.
- Authorized staff members and Business Owner will maintain and update business data such as customer profiles, supplier records, and menu information regularly.

## 5 Use Cases

### 5.1 Use Case Matrix

| Use Case #                            | Use Case Name           | Actor(s)                             | Trigger                                                                      | Outcome                                                                                    | Priority [0 – 100/highest] |
|---------------------------------------|-------------------------|--------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------|
| <b>Payroll &amp; Staff Management</b> |                         |                                      |                                                                              |                                                                                            |                            |
| <b>US-PS-01</b>                       | Register New Staff      | Payroll Admin                        | Admin enters new staff details (Name, Contact Info, Role).                   | Staff record is saved in the system.                                                       | 95                         |
| <b>US-PS-02</b>                       | Update Staff Profile    | Payroll Admin                        | Staff details change (e.g., phone number or role).                           | Staff record is successfully modified.                                                     | 85                         |
| <b>US-PS-03</b>                       | Deactivate Staff        | Business Owner                       | Staff member leaves the company.                                             | Staff is marked "Inactive" (history is preserved).                                         | 75                         |
| <b>US-PS-04</b>                       | Record Staff Payment    | Payroll Admin                        | Admin specifies amount, date, and payment method.                            | Payment record is stored for the employee.                                                 | 100                        |
| <b>US-PS-05</b>                       | View Payment History    | Payroll Admin                        | Admin selects a staff member and date range.                                 | Payment history is displayed for review.                                                   | 80                         |
| <b>US-PS-06</b>                       | Generate Payroll report | Payroll Admin                        | Admin requests a monthly summary.                                            | A monthly payroll report is generated.                                                     | 80                         |
| <b>User Management</b>                |                         |                                      |                                                                              |                                                                                            |                            |
|                                       | Register User Account   | Business Owner, Authorized Personnel | A new staff member requires system access to perform their duties.           | A new user account is created with system access privileges and saved in the database.     | 100                        |
| <b>US-UM-01</b>                       | User Login              | All Staffs                           | User enters staff ID and password on the login page and submits the request. | The user successfully logs into the system and is redirected to the appropriate dashboard. | 100                        |
| <b>US-UM-02</b>                       | User Logout             | All Staffs                           | User selects the logout option from the system.                              | The system ends the current session securely and returns                                   | 90                         |



|                            |                             |                                      |                                                                                                  |                                                                                                      |     |
|----------------------------|-----------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----|
|                            |                             |                                      |                                                                                                  | the user to the login page.                                                                          |     |
|                            | Update User Account         | Business Owner, Authorized Personnel | A user's role or access requirements change, or they need to update account details.             | The user account details or permissions are successfully updated in the database.                    | 85  |
|                            | Deactivate User Account     | Business Owner, Authorized Personnel | An employee leaves the company or changes to a role that no longer requires system access.       | The user account status is changed to "Inactive", preventing future logins while preserving history. | 90  |
| <b>Customer Management</b> |                             |                                      |                                                                                                  |                                                                                                      |     |
| <b>US-CM-01</b>            | Create Customer Profile     | Store Staff                          | A new customer requests catering services and does not exist in the system.                      | A new customer profile is created and stored securely.                                               | 90  |
| <b>US-CM-02</b>            | Update Customer Profile     | Store Staff                          | An existing customer requests changes to their contact information.                              | The customer profile is updated successfully.                                                        | 85  |
| <b>US-CM-03</b>            | View Customer Profile       | Store Staff                          | Staff needs to review customer details.                                                          | The customer profile is displayed to authorized users.                                               | 80  |
|                            | View Customer Report        | Business Owner, Store Staff          | The Business Owner or Store Staff needs to analyze customer behaviors or review business trends. | A formatted customer report is generated and displayed on the screen for review.                     | 85  |
| <b>US-CM-04</b>            | Deactivate Customer Profile | Business Owner                       | A customer becomes inactive or no longer uses the service.                                       | The customer profile is marked as inactive and no longer available for active operations.            | 75  |
| <b>Catering Operations</b> |                             |                                      |                                                                                                  |                                                                                                      |     |
| <b>US-CO-01</b>            | Create Catering Order       | Store Staff                          | A customer requests a new catering service.                                                      | A new order is saved with customer, menu, and event details.                                         | 100 |

|                                      |                                    |                |                                                                                 |                                                                                                                 |     |
|--------------------------------------|------------------------------------|----------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----|
| <b>US-CO-02</b>                      | Update Catering Order              | Store Staff    | A customer wants to change items, date, or event details.                       | The existing order is modified, and the system reflects new details.                                            | 90  |
| <b>US-CO-03</b>                      | Cancel Catering Order              | Store Staff    | A customer cancels their event.                                                 | The order is marked as "Cancelled" in the system.                                                               | 80  |
| <b>US-CO-04</b>                      | View Catering Orders               | Store Staff    | User wants to review upcoming catering events by date or event schedule.        | The system displays a list or calendar view of catering orders based on selected filters.                       | 80  |
|                                      | Process Website Catering Inquiries | Store Staff    | A new catering inquiry is submitted through the external Chocolatetown website. | A new Customer Profile and a Catering Order are created based on the inquiry details.                           | 95  |
| <b>US-CO-05</b>                      | Update Order Status                | Store Staff    | The kitchen starts preparing or the delivery is completed.                      | The order status is updated according to the order lifecycle (e.g., Confirmed → Preparing → Ready → Delivered). | 85  |
| <b>US-CO-06</b>                      | Generate Catering Invoice          | Store Staff    | A catering order is confirmed and needs billing.                                | A detailed invoice is created with auto-calculated total costs.                                                 | 100 |
| <b>US-CO-07</b>                      | Record Catering Payment            | Store Staff    | A customer makes a payment (deposit or balance).                                | The system records the payment (Unpaid/Partial/Paid ) and updates the balance.                                  | 100 |
| <b>Product &amp; Menu Management</b> |                                    |                |                                                                                 |                                                                                                                 |     |
| <b>US-PM-01</b>                      | Add New Menu Item                  | Business Owner | Owner needs to introduce a new dish or product.                                 | A new item (name, price, description) is created and available for orders.                                      | 95  |
| <b>US-PM-02</b>                      | Update Menu Item                   | Business Owner | Pricing or description needs to be modified.                                    | The item's details are updated while maintaining its history in the system.                                     | 85  |

|                            |                             |                      |                                                                                          |                                                                                                      |     |
|----------------------------|-----------------------------|----------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----|
| <b>US-PM-03</b>            | Deactivate Menu Item        | Business Owner       | An item is no longer offered or out of season.                                           | The item is marked as "Inactive" and hidden from the ordering menu, but its historical data remains. | 70  |
| <b>Supplier Management</b> |                             |                      |                                                                                          |                                                                                                      |     |
| <b>US-SM-01</b>            | Add New Supplier            | Supplier Coordinator | A new partnership is established with a vendor.                                          | A new supplier profile (contact, address, category) is created.                                      | 95  |
| <b>US-SM-02</b>            | Update Supplier Info        | Supplier Coordinator | A supplier changes their contact person, phone, or pricing.                              | The supplier's record is updated with the latest information.                                        | 90  |
| <b>US-SM-03</b>            | Deactivate Supplier         | Supplier Coordinator | A vendor is no longer used or fails to meet standards.                                   | The supplier is marked as "Inactive," preventing new orders while keeping history.                   | 75  |
| <b>US-SM-04</b>            | Record Supplier Transaction | Supplier Coordinator | The user records a new supplier transaction such as purchasing ingredients or materials. | Supplier transaction is saved in the system for tracking.                                            | 100 |
| <b>US-SM-05</b>            | Upload Supplier Invoice     | Supplier Coordinator | The user uploads or records a supplier invoice related to a transaction.                 | Supplier invoice is stored and linked to the transaction.                                            | 85  |
| <b>US-SM-06</b>            | Record Supplier Payment     | Supplier Coordinator | The user records a payment made to a supplier.                                           | Payment information is saved and linked to the supplier transaction.                                 | 95  |

## 5.2 Business Rules

| <b>Business Rule</b> | <b>Business Rule Description</b>                                                                                              | <b>Related UC</b> |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>BR01</b>          | Login shall only be successful when both a unique staffId and a valid password match a registered user account.               | US-UM-01          |
| <b>BR02</b>          | When a user logs out, the system shall terminate the active session immediately to prevent unauthorized reuse of the session. | US-UM-02          |

|             |                                                                                                                                                     |                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>BR03</b> | Each employee must be assigned a unique Employee ID within the system upon registration.                                                            | US-PS-01                                        |
| <b>BR04</b> | Staff profile information may only be updated by the Business Owner or authorized Payroll Administrator.                                            | US-PS-02                                        |
| <b>BR05</b> | Staff payment records must include the amount, date, and payment method before they can be saved.                                                   | US-PS-04                                        |
| <b>BR06</b> | Access to payroll and personal staff information is restricted to the Business Owner and authorized Payroll Administrators.                         | US-PS-02,<br>US-PS-04,<br>US-PS-05,<br>US-PS-06 |
| <b>BR07</b> | All payroll transactions must be retained and archived accurately to comply with CRA auditing standards.                                            | US-PS-04,<br>US-PS-05,<br>US-PS-06              |
| <b>BR08</b> | Customer profiles must include valid contact information to ensure communication for catering events.                                               | US-CM-01,<br>US-CM-02                           |
| <b>BR09</b> | A catering order must be linked to a valid customer profile and contain at least one menu item before it can be created.                            | US-CO-01,<br>US-CO-02                           |
| <b>BR10</b> | Catering order status may only follow valid progression defined by the business (e.g., Pending, Confirmed, Preparing, Ready, Delivered, Cancelled). | US-CO-03,<br>US-CO-05                           |
| <b>BR11</b> | Once a catering order is marked as "Delivered" or "Cancelled", it can no longer be modified.                                                        | US-CO-02,<br>US-CO-03,<br>US-CO-05              |
| <b>BR12</b> | The system shall use the pricing captured at the time of order creation for invoice generation.                                                     | US-CO-01<br>US-CO-06                            |
| <b>BR13</b> | Catering payment records must include the payment amount, payment date, and payment method before they can be saved.                                | US-CO-07                                        |
| <b>BR14</b> | All catering payments must be associated with a valid catering invoice.                                                                             | US-CO-06,<br>US-CO-07                           |
| <b>BR15</b> | Invoice payment status must be automatically updated based on the total payments received (e.g., Unpaid, Partially Paid, Fully Paid).               | US-CO-06,<br>US-CO-07                           |
| <b>BR16</b> | Menu pricing can only be modified by the Business Owner, and these changes shall not affect previously generated catering invoices.                 | US-PM-02,<br>US-CO-06                           |
| <b>BR17</b> | Each supplier must have a unique supplier ID in the system                                                                                          | US-SM-01                                        |
| <b>BR18</b> | A supplier transaction must include the supplier, transaction date, and amount before it can be saved                                               | US-SM-04                                        |
| <b>BR19</b> | A supplier invoice must be linked to an existing supplier transaction before it can be recorded                                                     | US-SM-05                                        |
| <b>BR20</b> | A supplier payment record must include the payment amount, date, and method before it can be saved                                                  | US-SM-06                                        |
| <b>BR21</b> | Deactivated staff, customers, menu items, and suppliers must remain in the system for historical tracking and auditing purposes.                    | US-PS-03,<br>US-CM-04,<br>US-PM-03,<br>US-SM-03 |

|             |                                                                                                             |                                                                                        |
|-------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>BR22</b> | Inactive staff, customers, menu items, and suppliers cannot be selected for new transactions or operations. | US-PS-03,<br>US-CM-04,<br>US-CO-01,<br>US-CO-02,<br>US-PM-03,<br>US-SM-03,<br>US-SM-04 |
|-------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

## 5.3 Use Cases

### 5.3.1 US-PS-01: Register New Staff

#### 5.3.1.1 *User Story*

As a Business Owner or Payroll Admin,

I want to register new staff members by entering their name, contact info, and role, so that I can manage their profiles and process their payroll accurately.

#### 5.3.1.2 *Acceptance Criteria*

1. Given the Business Owner or Payroll Admin enters new staff details, then the system must validate that the name, contact info, and role are not empty.
2. Given the input data is valid, then the system must assign a unique Employee ID to the record.
3. Given a staff member with the same contact information already exists, then the system prevents duplicate creation and notifies the user.
4. Given the transaction is complete, then the system saves the staff profile to the database and displays a "Success" message.

### 5.3.1.3 UI Design (Mock up)

9:16

Chocolatetown ERP - Staff Management

Full Name (e.g., Jane Doe) \*

Phone Number (###-###-####) \*

Email Address (e.g., jane.doe@email.com)

Mailing Address

Role \*

Select Role

Hourly Rate (\$)

Save Profile

Clear Form

Go Back

Profile Preview



Name \*

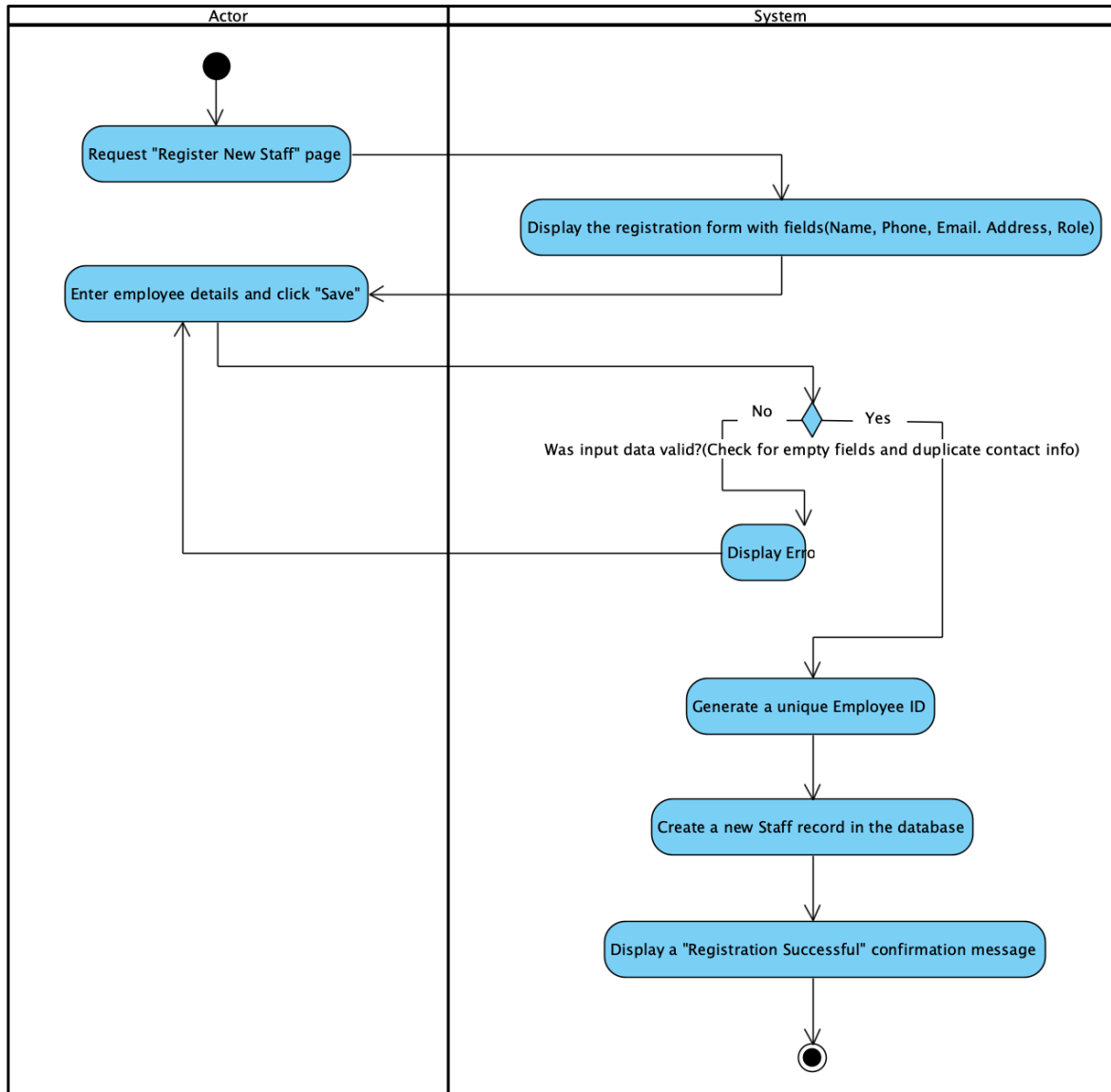
Phone Number (###-##)

Email Address

Mailing Address

### 5.3.1.4 Activity Diagram

act [UC03: Register New Staff]



### 5.3.1.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                            |                                                |                                                                                                           |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Use Case Name        | Register New Staff                                                                                                                                                                                                                                                                         |                                                |                                                                                                           |
| Triggering Event     | A new staff member joins Chocolatetown and needs a system account.                                                                                                                                                                                                                         |                                                |                                                                                                           |
| Brief Description    | Allows the Admin to enter personal and professional details for a new employee and save them as an active record.                                                                                                                                                                          |                                                |                                                                                                           |
| Actors               | Business Owner, Payroll Admin                                                                                                                                                                                                                                                              |                                                |                                                                                                           |
| Related Use Cases    | US-PS-04: Record Staff Payment (Payments are linked to registered staff)                                                                                                                                                                                                                   |                                                |                                                                                                           |
| Preconditions        | Admin has logged into the system.<br>Admin has appropriate permissions for staff management.                                                                                                                                                                                               |                                                |                                                                                                           |
| Post Conditions      | A new Staff object is instantiated, assigned a unique ID, and saved to the database with an "Active" status.                                                                                                                                                                               |                                                |                                                                                                           |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                      |                                                | System                                                                                                    |
|                      | 1.                                                                                                                                                                                                                                                                                         | Requests to register a new staff member.       | Displays the staff registration form.                                                                     |
|                      | 2.                                                                                                                                                                                                                                                                                         | Enters Name, Contact Info, and selects a Role. | Validates data, ensures no duplicates staff.<br>generates a unique Staff ID, set the default password.    |
|                      | 3.                                                                                                                                                                                                                                                                                         |                                                | Instantiates a new Staff object and saves it to the database.<br>Displays a success confirmation message. |
| Exception Conditions | <ul style="list-style-type: none"> <li>Missing Mandatory Data: If the name or role is missing, the system displays an error message preventing submission.</li> <li>Duplicate Entry: If the system detects a duplicate phone or email, it alerts the user and clears the input.</li> </ul> |                                                |                                                                                                           |

## 5.3.2 US-PS-02: Update Staff Profile

### 5.3.2.1 User Story



As a Business Owner or Payroll Admin,  
I want to update staff profile information (such as contact details or roles),  
so that employee records remain accurate and up-to-date for business operations.

#### *5.3.2.2 Acceptance Criteria*

1. Given the Admin selects an existing staff member, then the system must pre-fill the form with their current data.
2. Given the Admin modifies the information, then the system must validate that mandatory fields (Name, Role) are not empty.
3. Given the input data is valid and the "Save" button is clicked, then the system must update the specific Staff record in the database.
4. Given the update is successful, then the system displays a "Profile Updated Successfully" message.

### 5.3.2.3 UI Design (Mockup)

 Chocolatetown ERP

System > Staff > Update



## Edit Staff Profile

### Staff Member Details

**Staff ID** (Read-only text label)

[C7A-3489]

**Full Name:**

[Ethan Thorne]

**Phone Number:**

[416-555-4321]

**Email Address:**

[ethan.t@chocolatetown.net]

**Mailing Address:**

[100 Queen St. W, Ste 301]

**Role:**

[HR Specialist] ▼

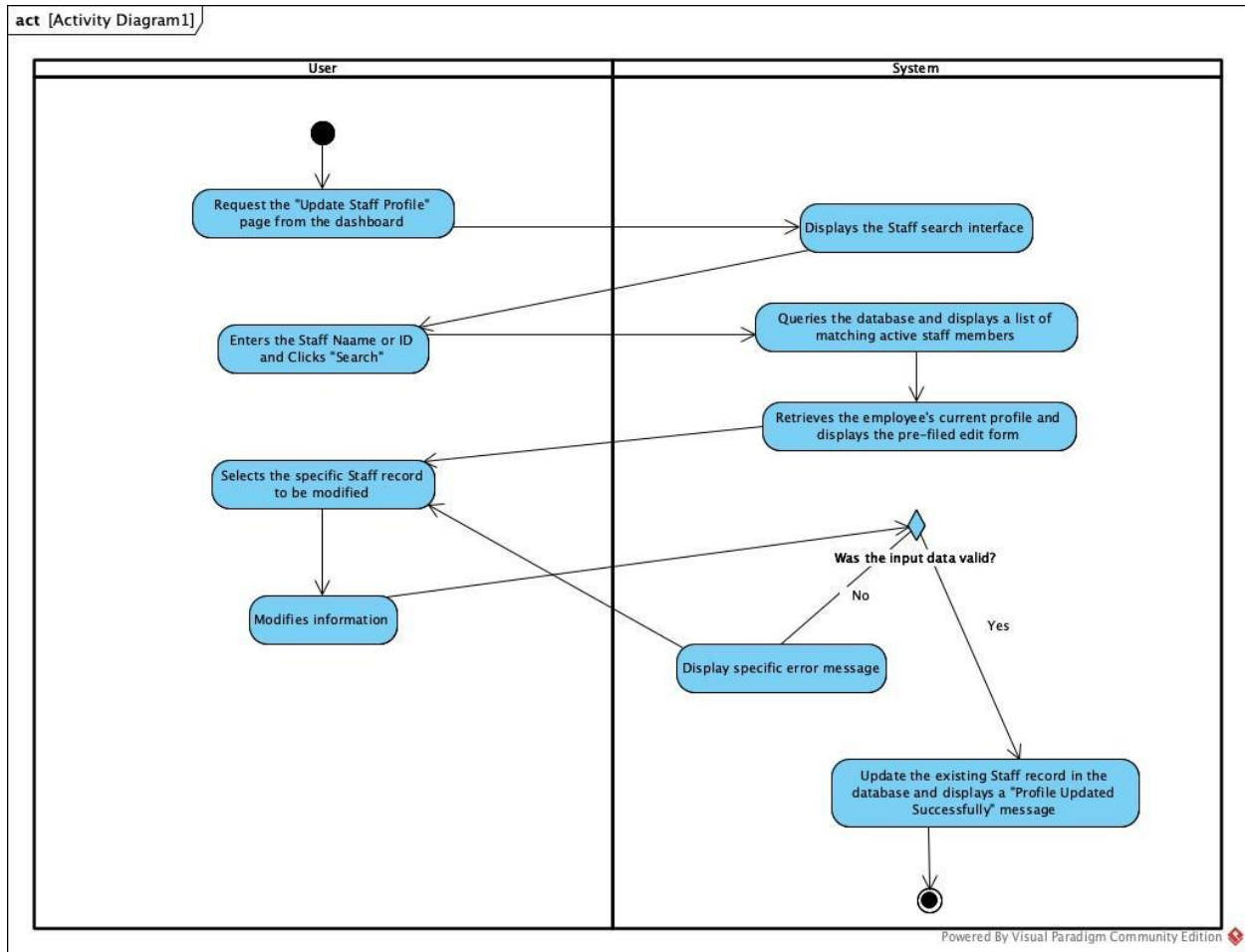
**Status:**

[Active] ▼

Update Profile

Cancel

### 5.3.2.4 Activity Diagram



### 5.3.2.5 Use Case Description

|                    |                                                                                                                                                                                                         |                                                                              |                                                          |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------|
| Use Case Name      | Update Staff Profile                                                                                                                                                                                    |                                                                              |                                                          |
| Triggering Event   | Staff details change or an error in the original record needs correction.                                                                                                                               |                                                                              |                                                          |
| Brief Description  | Allows the Admin to modify personal and professional details for an existing employee.                                                                                                                  |                                                                              |                                                          |
| Actors             | Payroll Admin                                                                                                                                                                                           |                                                                              |                                                          |
| Related Use Cases  | US-PS-01: Register New Staff (Target record must exist)<br>US-PS-03: Deactivate Staff (Profile update includes status change)<br>US-PS-04: Record Staff Payment (Payments are linked to these profiles) |                                                                              |                                                          |
| Preconditions      | Admin has logged into the system.<br>The staff record exists in the system                                                                                                                              |                                                                              |                                                          |
| Post Conditions    | The Staff object in the database is modified with the new information.                                                                                                                                  |                                                                              |                                                          |
| Flow of activities | Actor                                                                                                                                                                                                   |                                                                              | System                                                   |
|                    | 1.                                                                                                                                                                                                      | Selects the "Update Staff" option and searches for an employee by Name or ID | Displays a list of matching staff members for selection. |

|                      |                                                                                                                                                                                                                                                                                                     |                                                                                  |                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                      | 2.                                                                                                                                                                                                                                                                                                  | Selects the specific employee to be updated.                                     | Retrieves the current profile from the database and pre-fills the edit form.                 |
|                      | 3.                                                                                                                                                                                                                                                                                                  | Modifies the contact information or role and clicks the "Update Profile" button. | I Validates that mandatory fields are not empty and ensures no duplicate email/phone exists. |
|                      | 4.                                                                                                                                                                                                                                                                                                  |                                                                                  | Saves the changes to the database and displays a "Profile Updated Successfully" message.     |
| Exception Conditions | <ul style="list-style-type: none"> <li>Missing Mandatory Data: If Name or Role is cleared, the system displays an error message preventing submission.</li> <li>Duplicate Entry: If the system detects a duplicate phone or email already assigned to another staff, it alerts the user.</li> </ul> |                                                                                  |                                                                                              |

### 5.3.3 US-PS-04: Record Staff Payment

#### 5.3.3.1 User Story

As a Payroll Admin,

I want to record staff payments by specifying the amount, date, and payment method, so that Chocolatetown maintains accurate financial records for CRA auditing.

#### 5.3.3.2 Acceptance Criteria

1. Given the Payroll Admin selects an active staff member, then the system retrieves their current profile.
2. Given the Payroll Admin enters a payment, then the system validates that the amount, date, and payment method are provided.
3. Given the payment amount is zero or negative, then the system prevents saving and displays an error message.
4. Given the payment record is valid, then the system must link the transaction to the specific Staff object.
5. Given the transaction is saved, then the system stores the data for future payroll reports and audit tracking.

### 5.3.3.3 UI Design (Mockup)

9:16

Chocolatetown ERP - Payroll & Finance

Payment Amount \*

Payment Amount \*

Role \*

Select Role

▼

Payment Amount (\$)

Submit Payment

Reset

Go Back

Payment Preview



Name \*

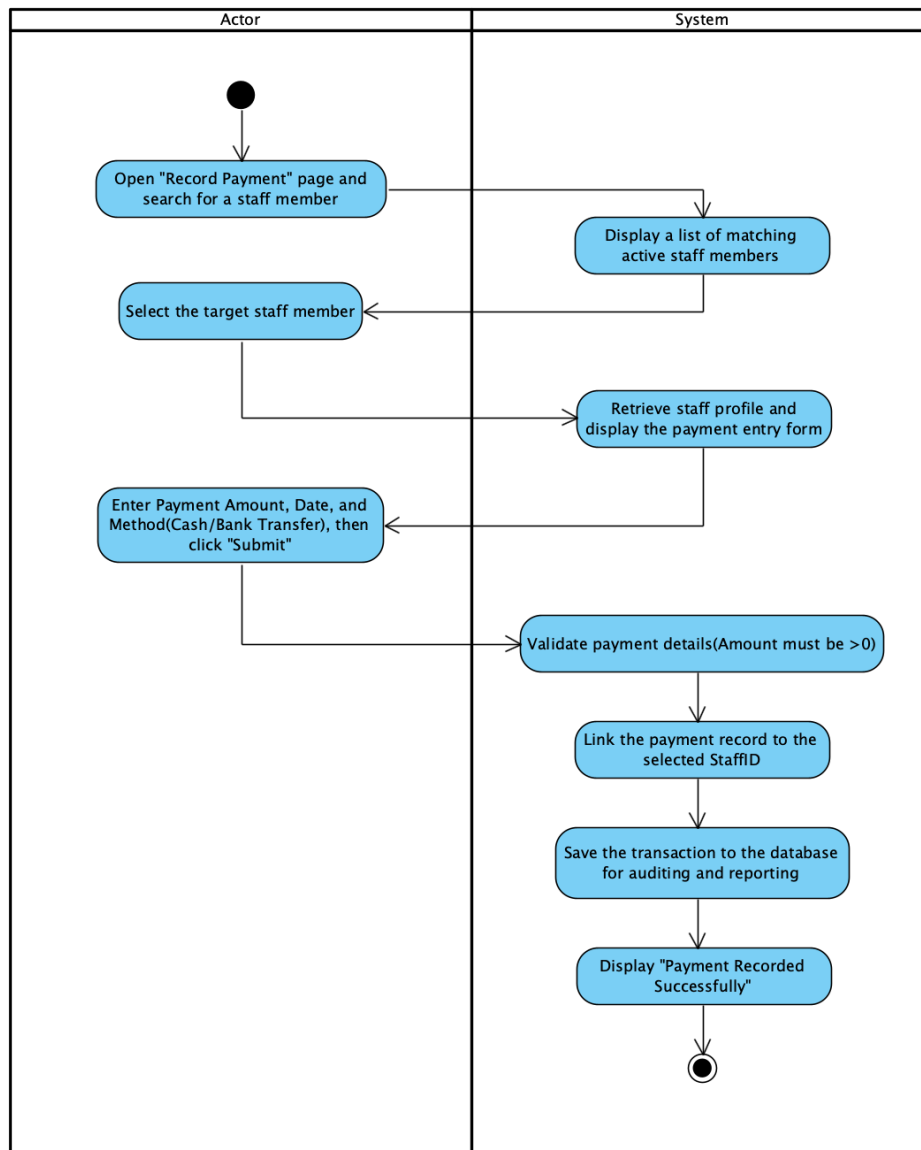
Phone Number (###-##)

Email Address

Mailing Address

### 5.3.3.4 Activity Diagram

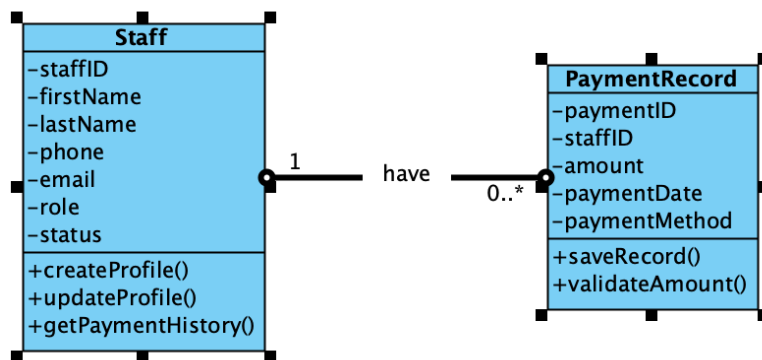
act [UC04: Record Staff Payment]



### 5.3.3.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                          |                                                                              |                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name        | Record Staff Payment                                                                                                                                                                                                                                                                                                     |                                                                              |                                                                                                                                      |
| Triggering Event     | A salary or hourly payment has been issued to an employee.                                                                                                                                                                                                                                                               |                                                                              |                                                                                                                                      |
| Brief Description    | Allows the Payroll Admin to document individual staff payments to ensure organized payroll tracking and legal compliance.                                                                                                                                                                                                |                                                                              |                                                                                                                                      |
| Actors               | Payroll Admin                                                                                                                                                                                                                                                                                                            |                                                                              |                                                                                                                                      |
| Related Use Cases    | US-PS-01: Register New Staff (Pre-requisite)<br>US-PS-06: Generate Payroll Report (Uses data from this UC)                                                                                                                                                                                                               |                                                                              |                                                                                                                                      |
| Preconditions        | Admin has logged into the system.<br>The staff member is registered and marked as "Active" in the system.                                                                                                                                                                                                                |                                                                              |                                                                                                                                      |
| Post Conditions      | A new PaymentRecord object is instantiated and saved to the database, linked to the Staff member.                                                                                                                                                                                                                        |                                                                              |                                                                                                                                      |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                                    |                                                                              | System                                                                                                                               |
|                      | 1.                                                                                                                                                                                                                                                                                                                       | Selects the option to record a staff payment.                                | Displays a staff search field and payment entry form.                                                                                |
|                      | 2.                                                                                                                                                                                                                                                                                                                       | Selects the target employee and enters the Amount, Date, and Payment Method. | Validates that the amount is positive and all mandatory fields are filled.                                                           |
|                      | 3.                                                                                                                                                                                                                                                                                                                       |                                                                              | Instantiates a PaymentRecord object linked to the employee. Saves the record and displays a "Payment Recorded Successfully" message. |
| Exception Conditions | <ul style="list-style-type: none"> <li>Missing Mandatory Data: If amount, date, or method is missing, the system highlights the field and prevents submission.</li> <li>Inactive Staff: If the user attempts to record a payment for an inactive staff member, the system displays a warning (Based on BR22).</li> </ul> |                                                                              |                                                                                                                                      |

### 5.3.3.6 Class Diagram





### 5.3.4 US-PS-05: View Payment History

#### 5.3.4.1 *User Story*

As a Payroll Admin,

I want to view the payment history of a staff member for a selected period, so that I can review past compensation records for auditing and financial tracking.

#### 5.3.4.2 *Acceptance Criteria*

1. Given the Admin selects a staff member, then the system must retrieve all associated payment records.
2. Given the Admin specifies a date range, then the system must filter the results to display only payments within that period.
3. Given the results are displayed, then the system must show the Date, Amount, and Payment Method for each transaction.
4. Given no records exist for the selected period, then the system must display a message "No payment records found."

### 5.3.4.3 UI Design (Mockup)

 Chocolate Town ERP



System > Payroll > Payment History

## Staff Payment History

### Staff Payment History

Search Staff ID/Name...

Date Range:  to

View History

### Recent Payments

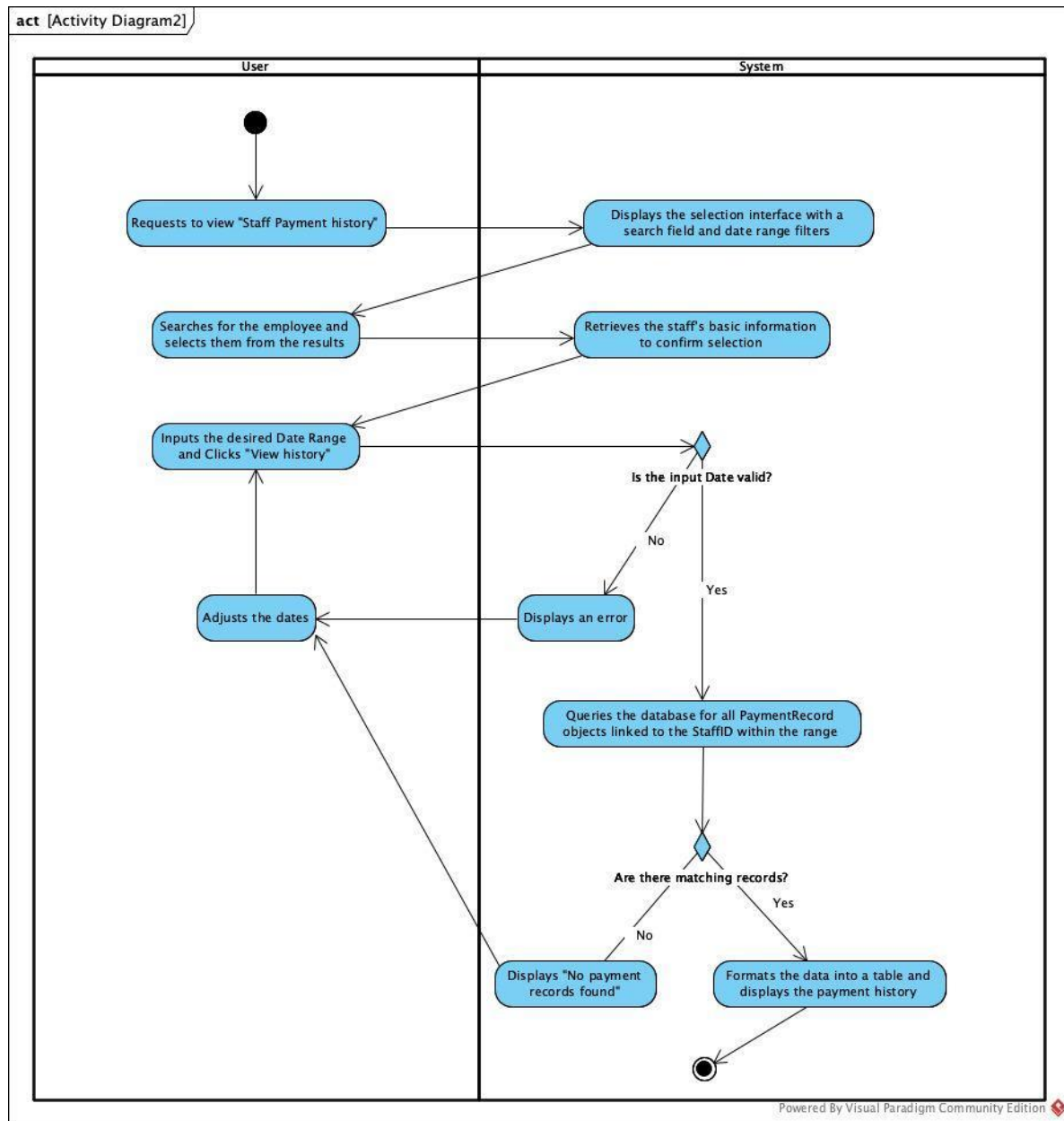
| Date                | Amount     | Method             |
|---------------------|------------|--------------------|
| Date:<br>2026/03/01 | [\$500.00] | [Bank<br>Transfer] |
| Date:<br>2026/03/15 | [\$550.00] | [Cash]             |

<< < 1 > >>

Export to PDF

Back

### 5.3.4.4 Activity Diagram

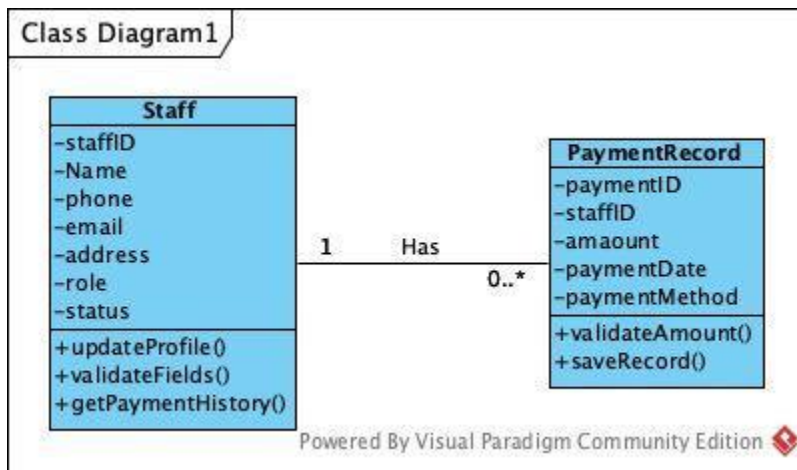


### 5.3.4.5 Use Case Description

|                   |                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name     | View Payment History                                                                                                                |
| Triggering Event  | Admin needs to review or verify past payments for a specific employee.                                                              |
| Brief Description | Allows the Admin to filter and display all historical payment transactions for a selected staff member within a specific timeframe. |
| Actors            | Payroll Admin                                                                                                                       |

|                      |                                                                                                                                                                                                                                                                                                       |                                                                               |                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Related Use Cases    | US-PS-04: Record Staff Payment (Provides the data for history)<br>US-PS-06: Generate Payroll Report (Summarizes this historical data)                                                                                                                                                                 |                                                                               |                                                                                                       |
| Preconditions        | Admin has logged into the system.<br>The selected staff member is registered in the system.                                                                                                                                                                                                           |                                                                               |                                                                                                       |
| Post Conditions      | Payment data is retrieved from the database and displayed in a formatted table.                                                                                                                                                                                                                       |                                                                               |                                                                                                       |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                 | System                                                                        |                                                                                                       |
|                      | 1.                                                                                                                                                                                                                                                                                                    | Requests to view the payment history for a staff member.                      | Displays the staff selection menu and date range filter fields (Start Date/End Date).                 |
|                      | 2.                                                                                                                                                                                                                                                                                                    | Selects the target employee and enters the desired date range for the review. | Queries the database for all PaymentRecord objects linked to the selected StaffID within that period. |
|                      | 3.                                                                                                                                                                                                                                                                                                    | Clicks the "View History" button to generate the list.                        | Retrieves the filtered data and displays the payment history in a formatted table.                    |
| Exception Conditions | <ul style="list-style-type: none"> <li>No Records Found: If no payments are found for the criteria, the system displays a "No payment records found" notification.</li> <li>Invalid Date Range: If the start date is after the end date, the system prompts the user to correct the range.</li> </ul> |                                                                               |                                                                                                       |

#### 5.3.4.6 Class Diagram



### 5.3.5 US-CM-01: Create Customer Profile

| Use Case ID | Use Case                | Use Case Description                                                       | Trigger                                                                     | Actor       | Brief Description                                                                                                                                                                                             |
|-------------|-------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-CM-01    | Create Customer Profile | Store Staff should be able to create a new customer profile in the system. | A new customer requests catering services and does not exist in the system. | Store Staff | The actor enters customer information such as name and contact details. The system validates the input, ensures required fields are complete and valid, and then creates and stores the new customer profile. |

#### 5.3.5.1 User Story

As a Store Staff,

I want to create a new customer profile when a customer requests catering services, so that I can store their information and manage future catering orders efficiently.

#### 5.3.5.2 Acceptance Criteria

1. Given the customer does not already exist in the system, When the Store Staff chooses to create a new customer profile, Then the system shall display a form for entering customer information.
2. Given a customer with the same identifying information already exists in the system, When the Store Staff attempts to create a duplicate customer profile, Then the system shall prevent duplicate creation and notify the user.
3. Given the Store Staff is submitting customer information, When the contact information entered is invalid, Then the system shall prompt the user to correct the invalid input before submission.
4. Given the Store Staff is submitting a new customer profile, When any required contact information is missing, Then the system shall prevent submission and display an error message indicating the missing field(s).
5. Given the Store Staff is entering customer information, When all required fields are completed with valid data, Then the system shall create and store the new customer profile successfully.

### 5.3.5.3 UI Design (Mockup)

ooo

## Customer Profile

CUSTOMER ID: 2026013557

NAME

Jlara Martins

PHONE

226-123-4567

EMAIL

hello@reallygreatsite.com

ADDRESS

138 Eldora Ave, Toronto, On, Canada

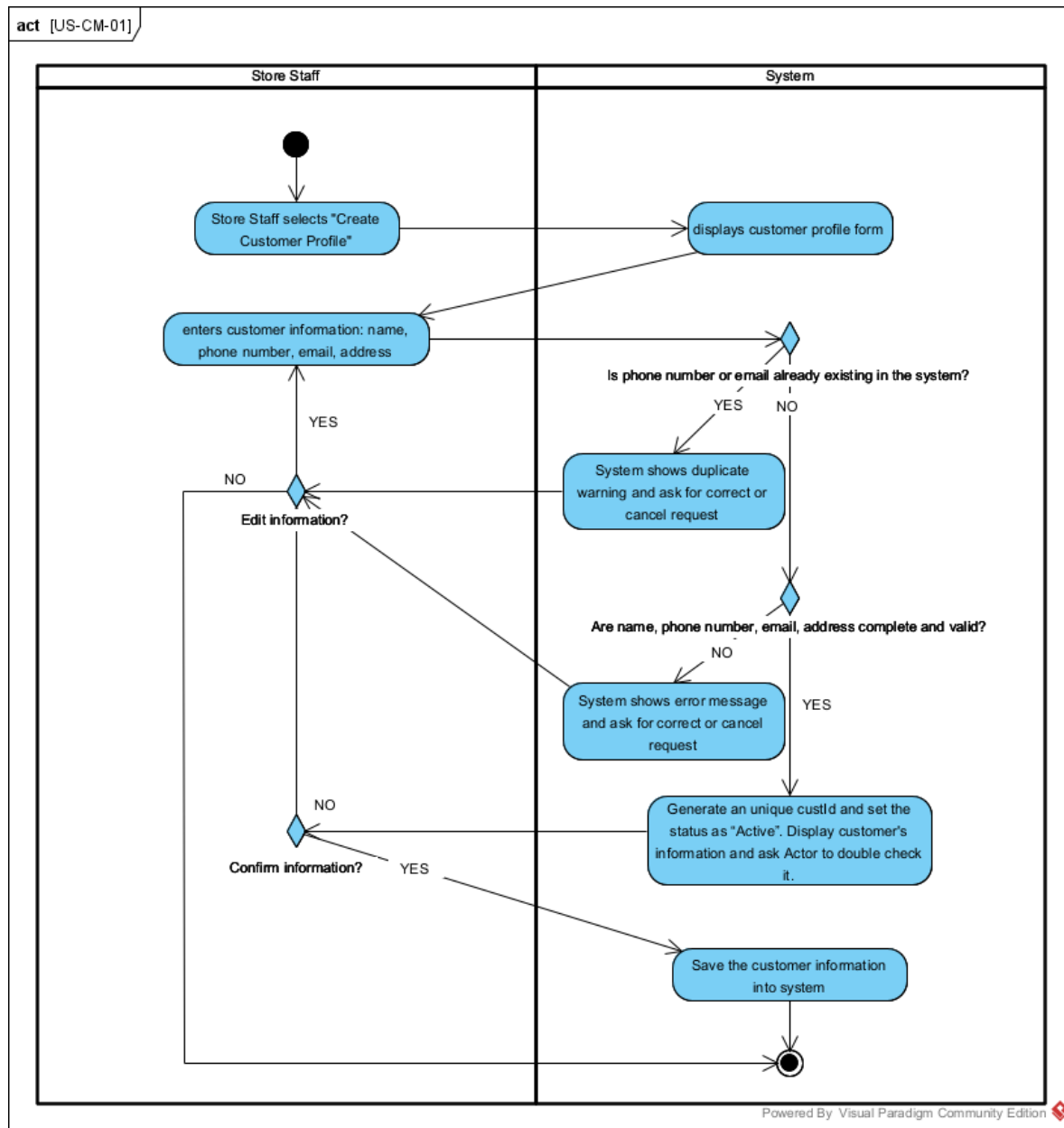
STATUS

Active

Save

Cancel

### 5.3.5.4 Activity Diagram

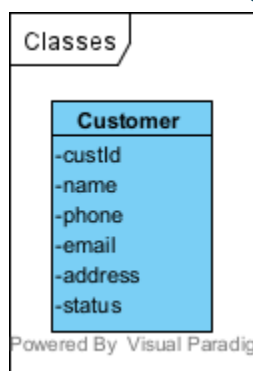


### 5.3.5.5 Use Case Description

|                   |                                                                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name     | Create Customer Profile                                                                                                                                                                                       |
| Triggering Event  | A new customer requests catering services and does not exist in the system.                                                                                                                                   |
| Brief Description | The actor enters customer information such as name and contact details. The system validates the input, ensures required fields are complete and valid, and then creates and stores the new customer profile. |
| Actors            | Store Staff                                                                                                                                                                                                   |

|                      |                                                                                                                                              |                                                                               |                                                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Related Use Cases    | US-CM-02 Update Customer Profile<br>US-CM-03 View Customer Profile<br>US-CM-04 Deactivate Customer Profile<br>US-CO-01 Create Catering Order |                                                                               |                                                                                                                                                                                                                                              |
| Preconditions        | The user is logged into the system.<br>The user has permission to create customer profiles.                                                  |                                                                               |                                                                                                                                                                                                                                              |
| Post Conditions      | A new customer profile is created and stored in the system.<br>The profile is available for future catering orders and customer management.  |                                                                               |                                                                                                                                                                                                                                              |
| Flow of activities   | Actor                                                                                                                                        |                                                                               | System                                                                                                                                                                                                                                       |
|                      | 1.                                                                                                                                           | Request to create a new customer profile.                                     | Displays the customer profile creation form.                                                                                                                                                                                                 |
|                      | 2.                                                                                                                                           | Enters customer information such as name, contact number, email, and address. | Checks whether the contact number or email already exists in the system. Validates name, contact number, email, and address are complete and valid.<br>Generate a unique custId and set the status as "Active".<br>Prompts for confirmation. |
|                      | 3.                                                                                                                                           | Confirms submission.                                                          | Saves the customer profile securely into the system.                                                                                                                                                                                         |
| Exception Conditions | <ul style="list-style-type: none"> <li>The actor cancels the process</li> <li>System error</li> </ul>                                        |                                                                               |                                                                                                                                                                                                                                              |

#### 5.3.5.6 Class Diagram



#### 5.3.6 US-CM-02: Update Customer Profile

| Use Case ID | Use Case | Use Case Description | Trigger | Actor | Brief Description |
|-------------|----------|----------------------|---------|-------|-------------------|
|-------------|----------|----------------------|---------|-------|-------------------|



|          |                         |                                                                                              |                                                                     |             |                                                                                                                                                   |
|----------|-------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| US-CM-02 | Update Customer Profile | This use case allows authorized staff to modify existing customer information in the system. | An existing customer requests changes to their contact information. | Store Staff | The system retrieves the customer record, allows the staff to update the relevant fields, validates the input, and saves the updated information. |
|----------|-------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

#### *5.3.6.1 User Story*

As a Store Staff,

I want to update customer profile information,

so that customer records remain accurate and up to date, improving service quality and customer satisfaction.

#### *5.3.6.2 Acceptance Criteria*

1. Given the Store Staff accesses the customer management interface, When the system provides access to customer records, Then customer records are available for selection.
2. Given customer records are available in the system, When the Store Staff locates and selects a customer, Then the system displays the customer profile for editing.
3. Given the customer exists in the system, When the Store Staff updates the customer profile with valid contact information, Then the system saves the updated information successfully.
4. Given the customer exists in the system, When the Store Staff enters missing or invalid contact information, Then the system displays an error and does not save the changes.

#### *5.3.6.3 UI Design (Mockup)*

ooo

## Customer List

| Customer ID | Customer     | Phone        | Email                     | Profile              |
|-------------|--------------|--------------|---------------------------|----------------------|
| 2026013557  | John Smith   | 226-123-4567 | hello@reallygreatsite.com | <a href="#">EDIT</a> |
| 2026013558  | Alen Hunter  | 226-250-0209 | alen@gmail.com            | <a href="#">EDIT</a> |
| 2026013559  | Jamie Oliver | 613-526-1488 | JamieO@gmail.com          | <a href="#">EDIT</a> |

[back](#)

ooo

## Customer Profile

CUSTOMER ID: 2026013557

NAME

Jiara Martins

PHONE

226-123-4567

EMAIL

hello@reallygreatsite.com

ADDRESS

138 Eldora Ave, Toronto, On, Canada

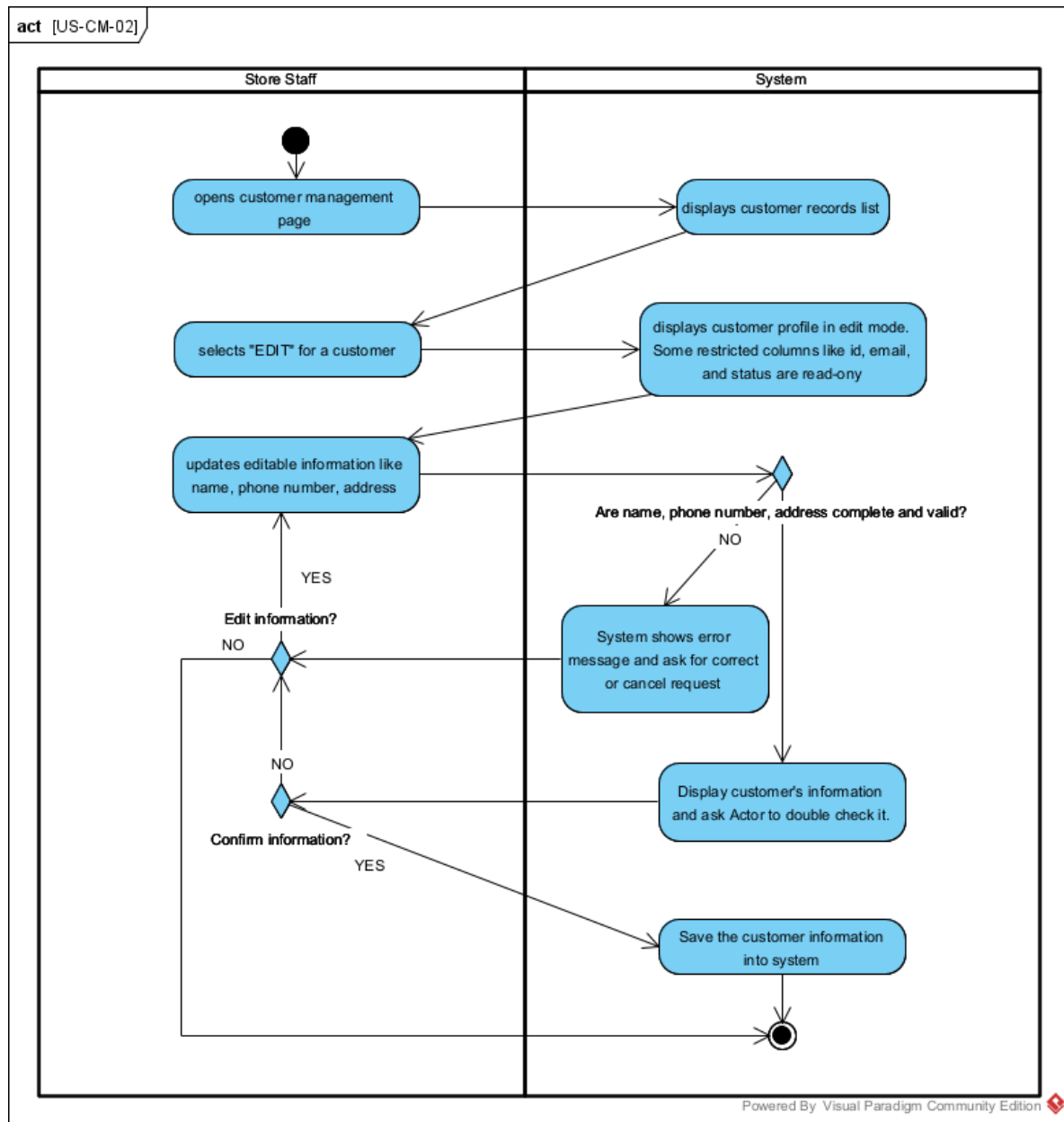
STATUS

Active

[UPDATE](#)

[Cancel](#)

### 5.3.6.4 Activity Diagram

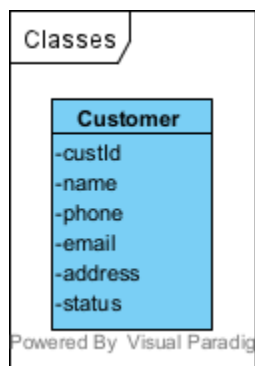


### 5.3.6.5 Use Case Description

|                   |                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name     | Update Customer Profile                                                                                                                           |
| Triggering Event  | An existing customer requests changes to their contact information.                                                                               |
| Brief Description | The system retrieves the customer record, allows the staff to update the relevant fields, validates the input, and saves the updated information. |
| Actors            | Store Staff                                                                                                                                       |

|                      |                                                                                                                                                                                                                   |                                                                        |                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Related Use Cases    | US-CM-01 Create Customer Profile<br>US-CM-03 View Customer Profile<br>US-CM-04 Deactivate Customer Profile<br>US-CO-01 Create Catering Order                                                                      |                                                                        |                                                                                                                                               |
| Preconditions        | The user is logged into the system.<br>The user has permission to update customer profiles.<br>The customer profile already exists in the system.                                                                 |                                                                        |                                                                                                                                               |
| Post Conditions      | The customer profile is updated and stored in the system.<br>The updated profile is available for future catering orders and customer management.<br>Restricted fields remain unchanged if they are not editable. |                                                                        |                                                                                                                                               |
| Flow of activities   | Actor                                                                                                                                                                                                             |                                                                        | System                                                                                                                                        |
|                      | 1.                                                                                                                                                                                                                | Requests access to customer records.                                   | Displays available customer records.                                                                                                          |
|                      | 2.                                                                                                                                                                                                                | Locates and selects a customer profile to edit.                        | Displays the selected customer profile in edit mode.<br>Shows editable fields and keeps restricted fields such as email and status read-only. |
|                      | 3.                                                                                                                                                                                                                | Updates allowed customer information such as phone number and address. | Validates the entered information.                                                                                                            |
|                      | 4.                                                                                                                                                                                                                | Confirms the update.                                                   | Saves the updated customer profile in the system.<br>Displays a confirmation that the profile has been updated successfully.                  |
| Exception Conditions | <ul style="list-style-type: none"> <li>• The actor cancels the process</li> <li>• System error</li> </ul>                                                                                                         |                                                                        |                                                                                                                                               |

### 5.3.6.6 Class Diagram



### 5.3.7 US-CO-01: Create Catering Order

| Use Case ID | Use Case              | Use Case Description                                                                                                       | Trigger                                     | Actor       | Brief Description                                                                                              |
|-------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------|
| US-CO-01    | Create Catering Order | As a Store Staff, I want to create a new catering order for a customer so that the business can fulfill the event request. | A customer requests a new catering service. | Store Staff | Store Staff selects a customer, chooses menu items, specifies event details, and creates a new catering order. |

#### 5.3.7.1 User Story

As a Store Staff.

I want to select a customer profile, choose menu items, and input event details to create a new catering order.

So that Chocolatetown can prepare and deliver the desserts for the customer's upcoming event.

#### 5.3.7.2 Acceptance Criteria

1. Given the Store Staff selects an existing customer, then the system retrieves and displays the customer's contact information.
2. Given the Store Staff adds menu items to the order, then the system automatically calculates the total cost based on current pricing.
3. Given the Store Staff submits the order, then the system validates that the event date is available and at least one menu item is selected.
4. Given the order is valid, then the system must instantiate a CateringOrder object and link it to the selected Customer profile.
5. Given the transaction is complete, then the system sets the order status to "Pending", saves the order, and displays a "Success" message.

### 5.3.7.3 UI Design (Mockup)

Chocolatetown Business Management System

Back

Profile

Create New Catering Order

1. Select Customer

John Wick (416-555-5555) ▾

Email: johnW@myseneca.com | Address: 123 seneca St.

2. Event Details

Event Date: 2026-03-30 | Time: 14:00 ▾

3. Menu Selection

[ v ] Premium Chocolate Fondue Set \$150.00 Qty: [ 1 ]

[ v ] Custom Soft Ice Cream Bar \$200.00 Qty: [ 1 ]

[ ] Assorted Chocolate Truffles \$ 50.00 Qty: [ 0 ]

Order Summary --

Subtotal: \$ 350.00

Tax: \$ 45.50

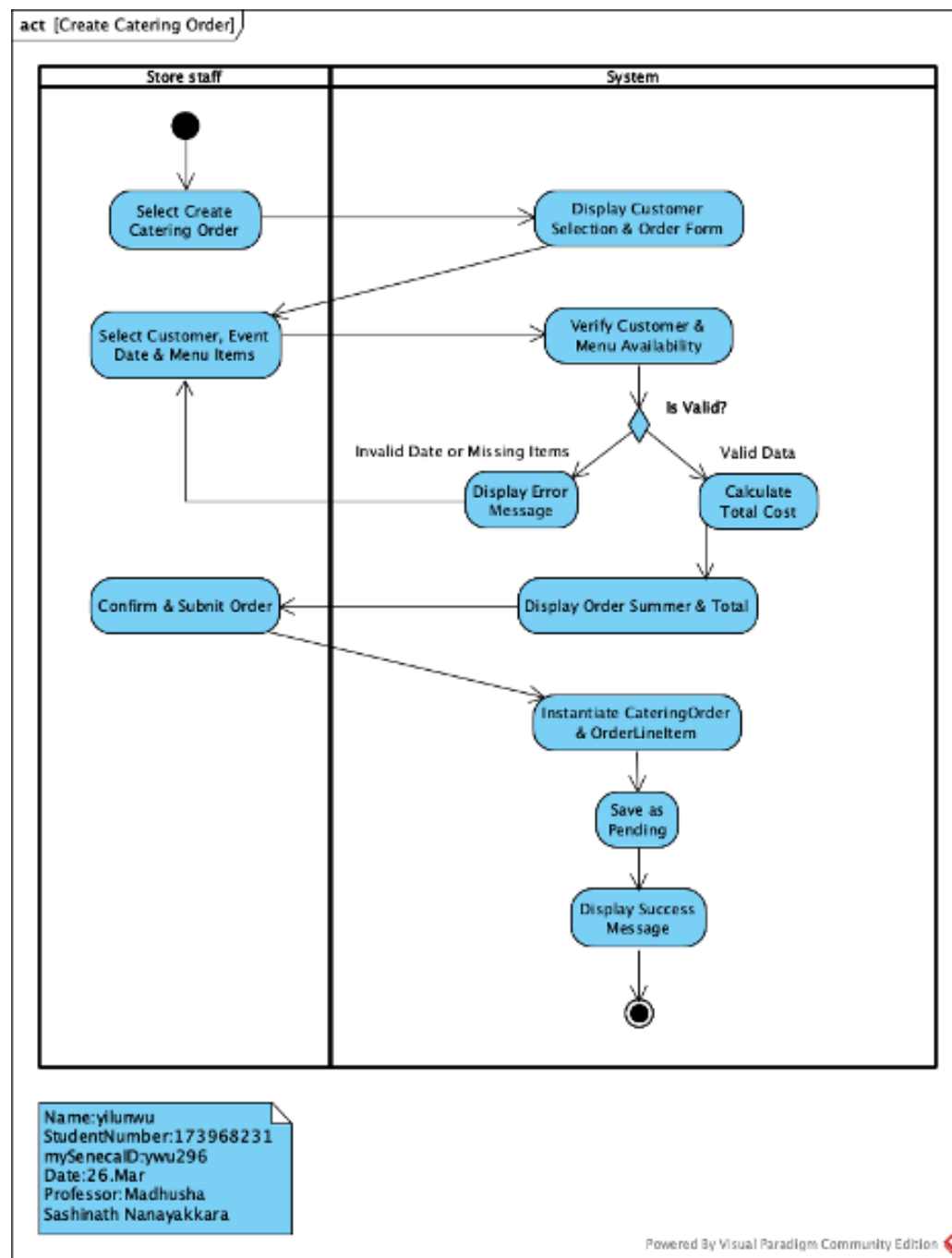
-----

Total: \$ 395.50

CANCEL

CREATE ORDER

### 5.3.7.4 Activity Diagram

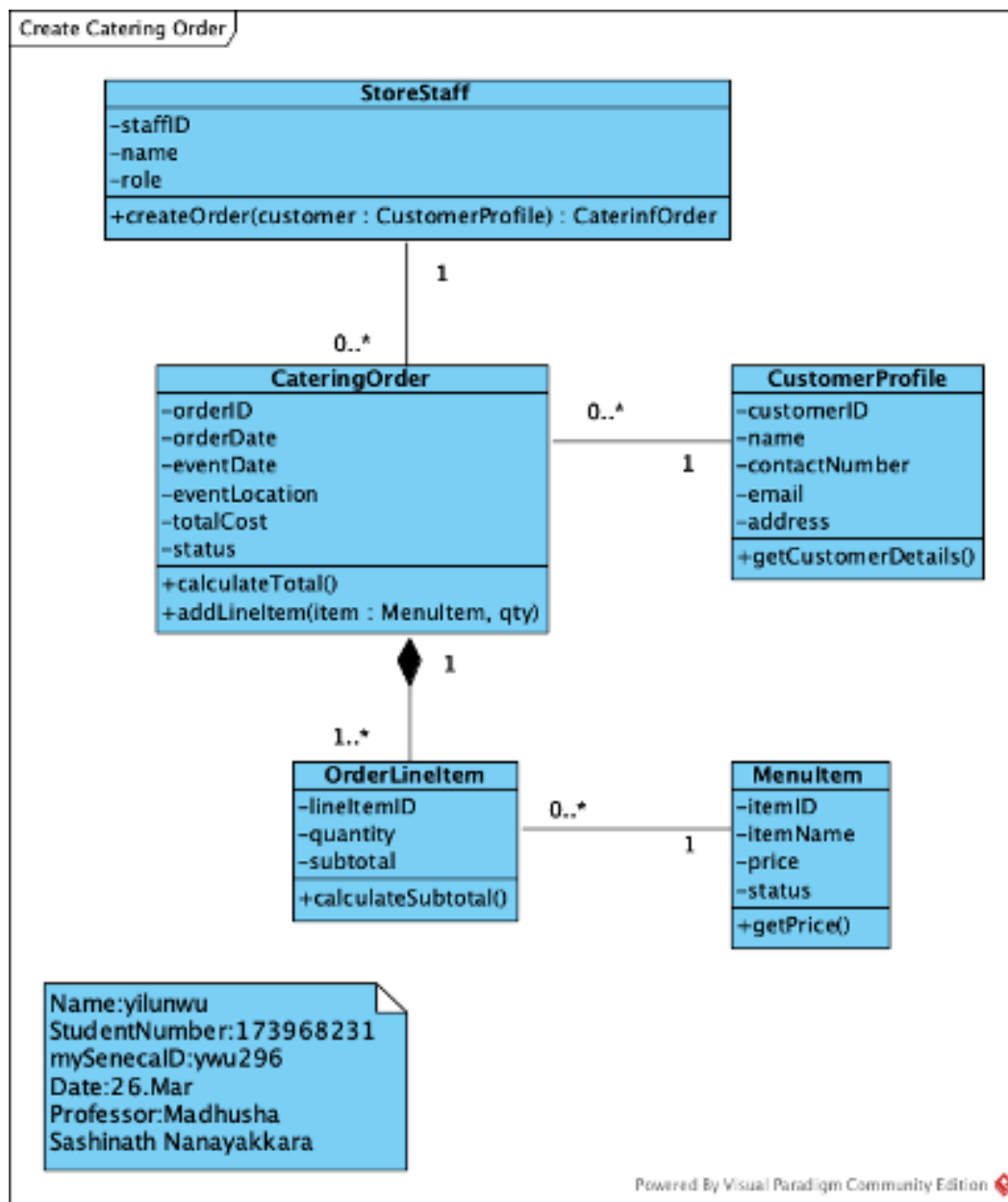


### 5.3.7.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                           |                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name        | Create Catering Order                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                           |                                                                                                                                              |
| Triggering Event     | A customer requests a new catering service for an upcoming event.                                                                                                                                                                                                                                                                                                                                            |                                                                                           |                                                                                                                                              |
| Brief Description    | Allows the Store Staff to select a customer profile, choose menu items, input event details, and securely save a new catering order in the system.                                                                                                                                                                                                                                                           |                                                                                           |                                                                                                                                              |
| Actors               | Store Staff                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                           |                                                                                                                                              |
| Related Use Cases    | US-CM-01: Create Customer Profile (Pre-requisite if the customer is new)<br>US-CO-06: Generate Catering Invoice (Often follows this step)                                                                                                                                                                                                                                                                    |                                                                                           |                                                                                                                                              |
| Preconditions        | 1. Store Staff has logged into the system.<br>2. The customer has an active profile in the system.<br>3. At least one active menu item exists in the database.                                                                                                                                                                                                                                               |                                                                                           |                                                                                                                                              |
| Post Conditions      | A new CateringOrder object is instantiated and saved to the database with a "Pending" status.                                                                                                                                                                                                                                                                                                                |                                                                                           |                                                                                                                                              |
| Flow of activities   | Actor (Store Staff)                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                           | System                                                                                                                                       |
|                      | 1.                                                                                                                                                                                                                                                                                                                                                                                                           | Requests to create a new catering order.                                                  | Displays the order creation form, prompting for customer selection, menu items, and event details.                                           |
|                      | 2.                                                                                                                                                                                                                                                                                                                                                                                                           | Selects the customer from the database.                                                   | Retrieves and auto-fills the customer's contact information.                                                                                 |
|                      | 3.                                                                                                                                                                                                                                                                                                                                                                                                           | Selects desired menu items, specifies quantities, and enters the event date and location. | Validates the input and automatically calculates the total cost based on the selected items and quantities.                                  |
|                      | 4.                                                                                                                                                                                                                                                                                                                                                                                                           | Confirms and submits the order request.                                                   | Instantiates a new CateringOrder object linked to the customer. Saves the order with a "Pending" status and displays a confirmation message. |
| Exception Conditions | <ul style="list-style-type: none"> <li>Missing Mandatory Data: If the order does not contain at least one menu item or is missing the event date, the system displays an error message preventing submission (Based on BR09).</li> <li>Inactive Profile/Item: If an inactive customer or discontinued menu item is selected, the system alerts the user and clears the selection (Based on BR22).</li> </ul> |                                                                                           |                                                                                                                                              |



### 5.3.7.6 Class Diagram



### 5.3.8 US-CO-02: Update Catering Order

| Use Case ID | Use Case              | Use Case Description                                                                                                                              | Trigger                                                                                                       | Actor       | Brief Description                                                                                                                                              |
|-------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-CO-02    | Update Catering Order | As a Store Staff, I want to modify the details of an existing catering order so that the system reflects the customer's most up-to-date requests. | A customer requests changes to their existing catering order (e.g., changing items, event date, or location). | Store Staff | The Store Staff retrieves an existing active order, updates the event details or menu items, and the system recalculates the total cost and saves the changes. |

#### 5.3.8.1 User Story

As a Store Staff.

I want to update the details of an existing catering order, such as modifying menu items, quantities, or the event date.

So that the kitchen prepares the correct items and the system maintains accurate order records for the customer.

#### 5.3.8.2 Acceptance Criteria

1. Given a catering order is currently marked as "Delivered" or "Cancelled", then the system must prevent the Store Staff from making any modifications to it.
2. Given the Store Staff selects an active catering order, then the system must retrieve and display the current order details and customer information.
3. Given the Store Staff updates the menu items or quantities, then the system must automatically recalculate the order's subtotal, tax, and total cost.
4. Given the Store Staff removes items from the order, then the system must validate that at least one valid menu item remains before allowing the update.
5. Given all updated information is valid and submitted, then the system saves the modifications to the database and displays a success confirmation message.

### 5.3.8.3 UI Design (Mockup)

Chocolatetown Business Management System

Back

Profile

Update Catering Order

1. Order Info

Order ID: **CT-ORD-1005** | Status: **Pending** | Date Created: 2026-03-15

2. Select Customer

John Wick (416-555-5555)

Email: johnW@myseneca.com | Address: 123 seneca St.

3. Event Details

Event Date: 2026-04-15 | Time: 14:00

4. Menu Selection

[ v ] Premium Chocolate Fondue Set \$150.00 Qty: [ 2 ] [DELETE]  
[ v ] Custom Soft Ice Cream Bar \$200.00 Qty: [ 1 ] [DELETE]  
[ ] Assorted Chocolate Truffles \$ 50.00 Qty: [ 0 ]

ADD ITEM

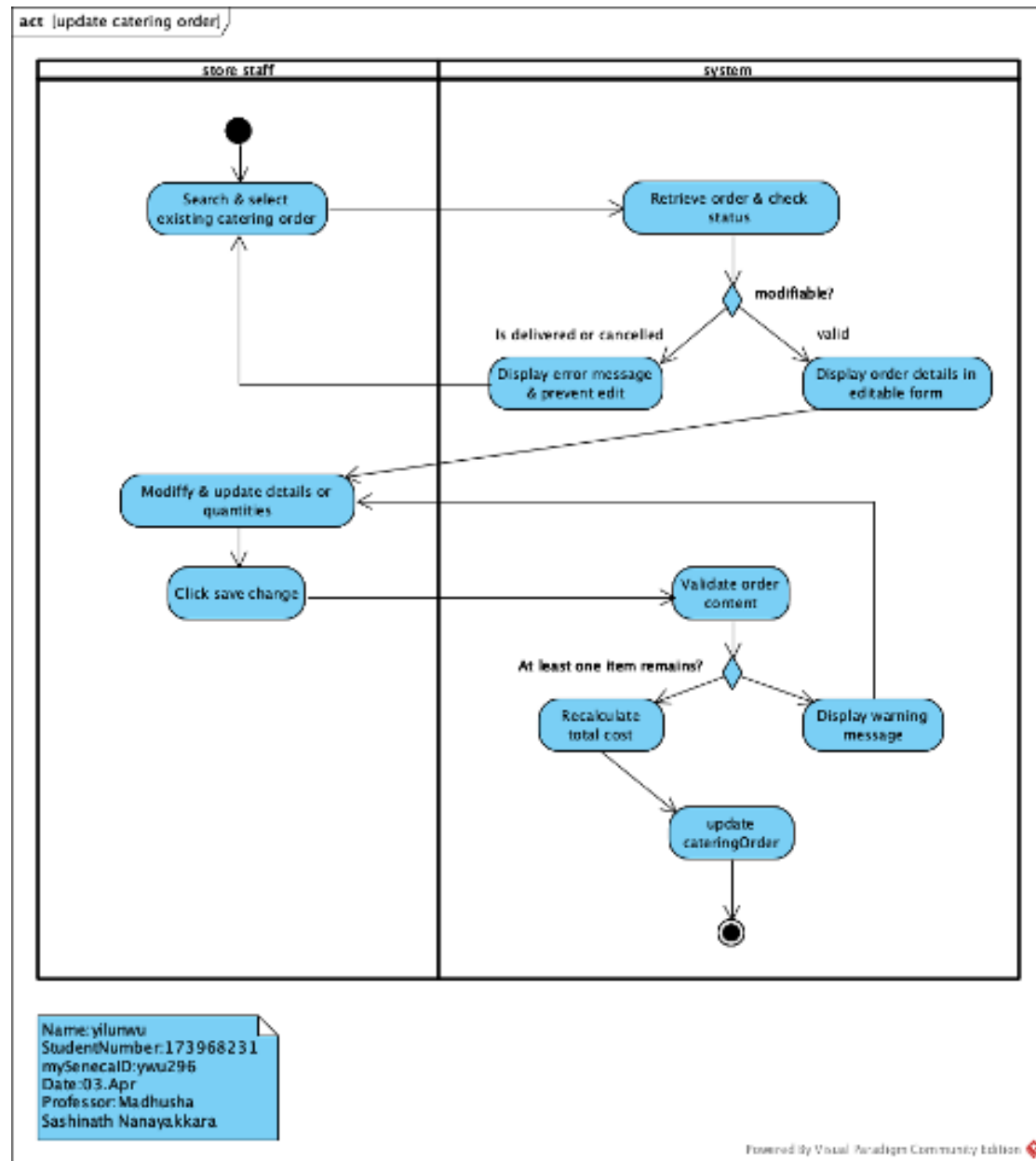
5. Order Summary

Subtotal: \$ 500.00  
Tax: \$ 65.00  
-----  
**Total: \$ 565.00**

CANCEL

SAVE CHANGES

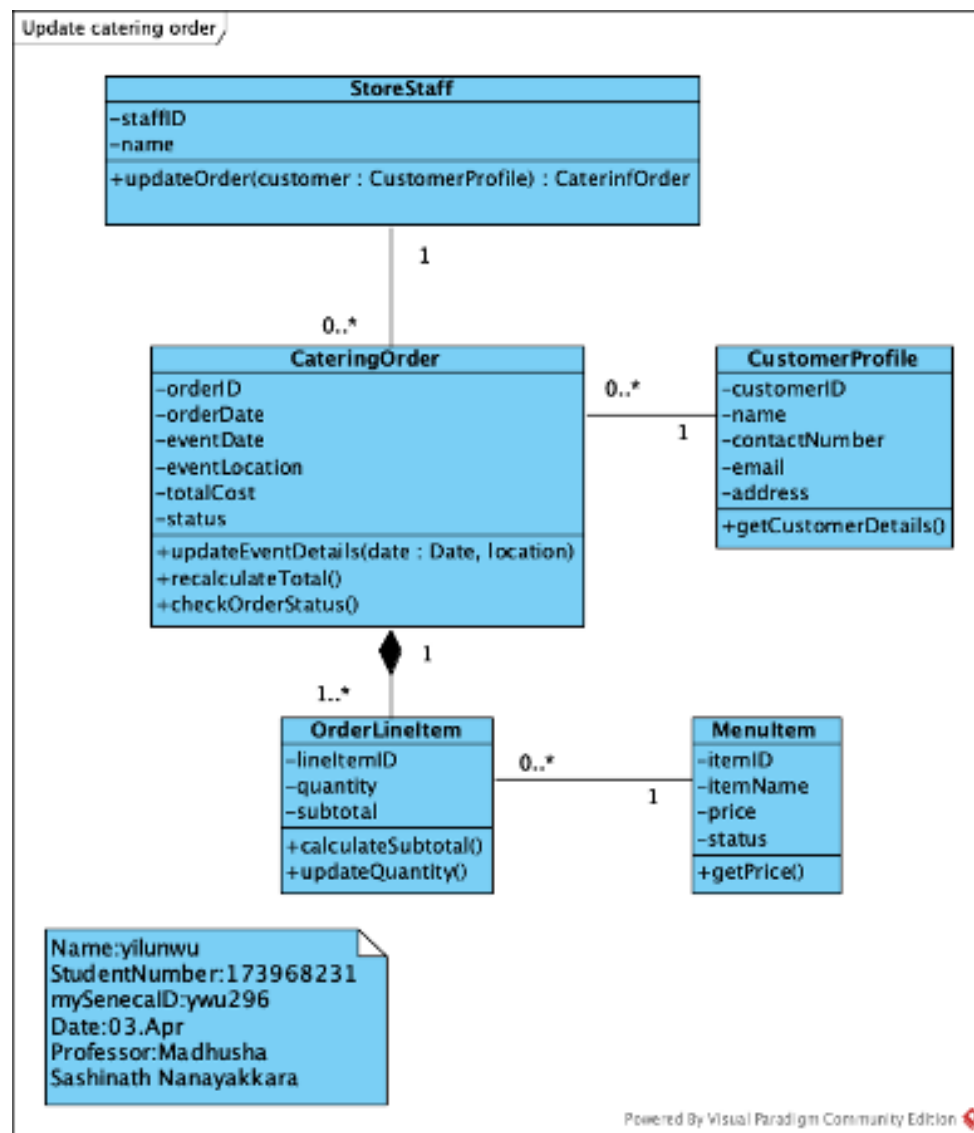
### 5.3.8.4 Activity Diagram



### 5.3.8.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                    |                                                                                                                                             |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name        | Update Catering Order                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                    |                                                                                                                                             |
| Triggering Event     | A customer contacts the store to request changes to their existing catering order (e.g., changing the date, adding or removing items).                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                    |                                                                                                                                             |
| Brief Description    | Allows the Store Staff to retrieve an active catering order, modify the event details or menu selections, automatically recalculate the order totals, and save the updated information to the database.                                                                                                                                                                                                                                                                                                    |                                                                                                                                                    |                                                                                                                                             |
| Actors               | Store Staff                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                    |                                                                                                                                             |
| Related Use Cases    | US-CO-01: Create Catering Order (Pre-requisite)<br>US-CO-06: Generate Catering Invoice (Affected by the updates)                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                    |                                                                                                                                             |
| Preconditions        | The Store Staff has successfully logged into the system.<br>The catering order exists in the system.<br>The order status is valid for modification (i.e., NOT "Delivered" or "Cancelled").                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                    |                                                                                                                                             |
| Post Conditions      | The existing CateringOrder object is successfully updated in the database. The total cost is recalculated, and the changes are reflected in the system.                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                    |                                                                                                                                             |
| Flow of activities   | Actor (Store Staff)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                    | System                                                                                                                                      |
|                      | 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Searches for and selects the existing catering order to update.                                                                                    | Verifies the order status. Retrieves and displays the current order details (customer info, event details, menu items) in an editable form. |
|                      | 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Modifies the event details (such as date, time, or location) and/or updates the menu items (adds new items, removes items, or changes quantities). | Validates the updated input. Ensures that at least one valid menu item remains in the order.                                                |
|                      | 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                    | Automatically recalculates the subtotal, tax, and total cost based on the updated menu selections and current pricing.                      |
|                      | 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Reviews the recalculated totals and submits the update request.                                                                                    | Updates the CateringOrder object in the database with the new details. Displays a "Success" confirmation message.                           |
| Exception Conditions | <ul style="list-style-type: none"> <li>Invalid Order Status: If the selected order is already marked as "Delivered" or "Cancelled", the system disables the editing features and displays an error message stating the order can no longer be modified (Based on BR11).</li> <li>No Menu Items Remaining: If the Store Staff removes all menu items from the order, the system prevents submission and displays a warning that a catering order must contain at least one item (Based on BR09).</li> </ul> |                                                                                                                                                    |                                                                                                                                             |

### 5.3.8.6 Class Diagram



### 5.3.9 US-CO-06: Generate Catering Invoice

| Use Case ID | Use Case                  | Use Case Description                                                                              | Trigger                                          | Actor       | Brief Description                                                                                                                                                                             |
|-------------|---------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-CO-06    | Generate Catering Invoice | The Store Staff generates a catering invoice for a confirmed catering order for billing purposes. | A catering order is confirmed and needs billing. | Store Staff | The actor selects a confirmed catering order to generate a detailed catering invoice. The system retrieves the order details, initializes the invoice payment tracking fields, and stores it. |

#### 5.3.9.1 User Story

As a Store Staff,

I want to generate a catering invoice for a confirmed order,

so that the customer can be billed accurately and the payment can be tracked properly.

#### 5.3.9.2 Acceptance Criteria

1. Given a catering order has been confirmed, When the Store Staff selects the option to generate an invoice, Then the system shall retrieve the order details and prepare the invoice.
2. Given a confirmed catering order contains selected menu items, When the invoice is generated, Then the system shall retrieve and use the pricing and cost stored in the catering order.
3. Given the invoice is being generated, When the system retrieves the stored order details successfully, Then the system shall display a detailed invoice showing the order items, quantities, unit prices, subtotals, and total amount.
4. Given an invoice has been generated, When no payment has yet been recorded, Then the invoice payment status shall be set to Unpaid.
5. Given an invoice has been generated successfully, When the invoice is saved, Then the system shall store the invoice.
6. Given menu prices are changed after an invoice has already been generated, When the user later views the existing invoice, Then the invoice shall retain the original prices used at the time it was generated.

5.3.9.3 UI Design (Mockup)

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Catering Order

| Order ID      | Customer     | Event Date | Order Status | Total Amount | Details | Invoice  | Payment Status |
|---------------|--------------|------------|--------------|--------------|---------|----------|----------------|
| CO-2026-00125 | John Smith   | 2026-04-20 | Confirmed    | \$ 181.93    | DETAILS | GENERATE | Unpaid         |
| CO-2026-00126 | Alen Hunter  | 2026-03-31 | Delivered    | \$ 257.66    | DETAILS | VIEW     | Partially Paid |
| CO-2026-00127 | Jamie Oliver | 2026-03-27 | Delivered    | \$ 95.64     | DETAILS | VIEW     | Fully Paid     |
| CO-2026-00127 | Jamie Oliver | 2026-03-27 | Cancelled    | \$ 95.64     | DETAILS |          |                |

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Invoice

INVOICE ID:CO-2026-00125-01

ORDER ID:CO-2026-00125EVENT DATE:2026-04-20  
CUSTOMER NAME:JOHN SMITHORDER STATUS:CONFIRMED

SELECTED ITEMS

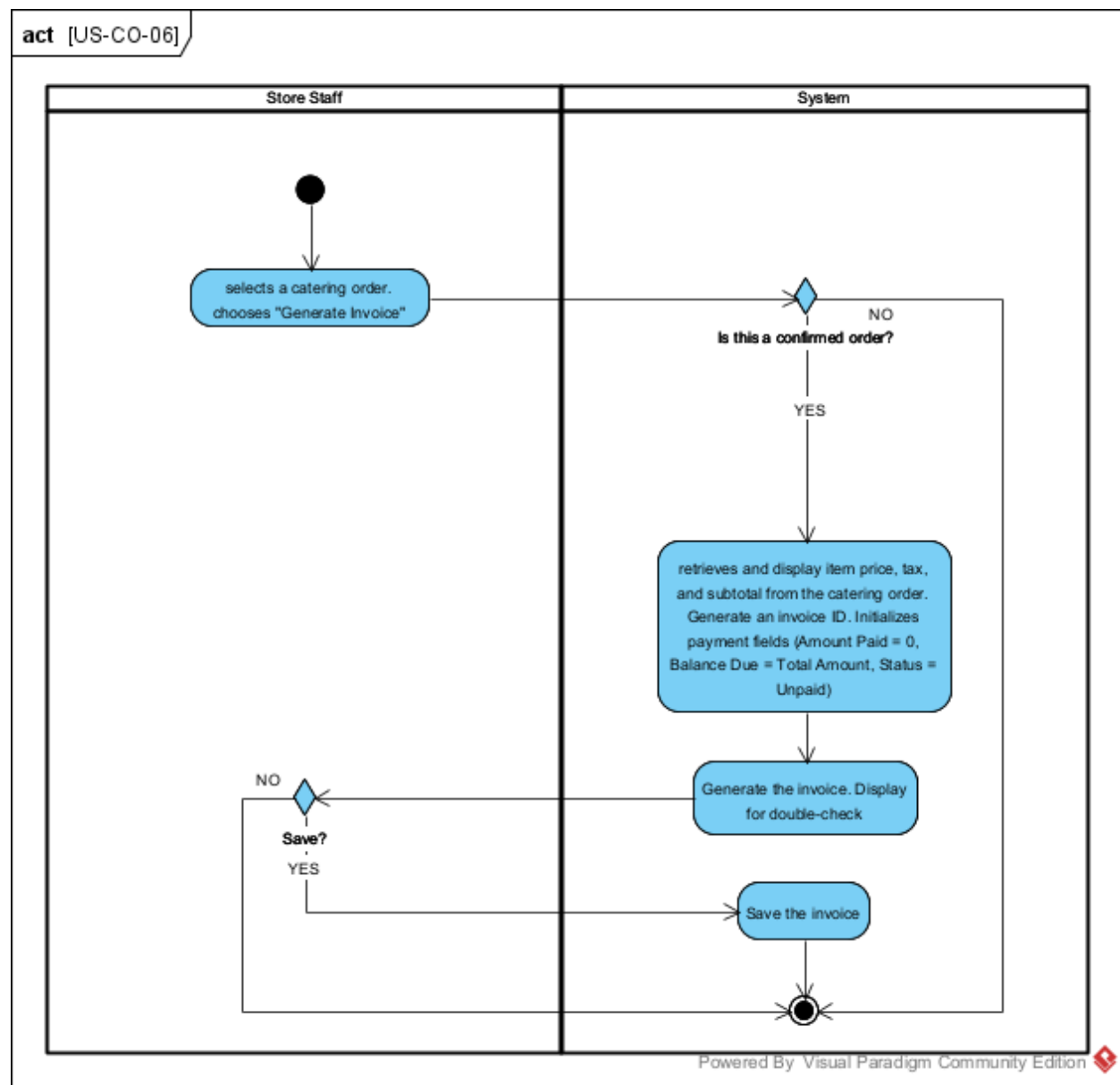
| Item Name      | Qty | Unit Price | Subtotal |
|----------------|-----|------------|----------|
| Sandwich Tray  | 2   | \$45.00    | \$90.00  |
| Fruit Platter  | 1   | \$35.00    | \$35.00  |
| Drinks Package | 3   | \$12.00    | \$36.00  |

SUBTOTAL:\$ 161.00  
TAX (13%)\$ 20.93  
TOTAL AMOUNT:\$ 181.93  
AMOUNT PAID:\$ 0.00  
BALANCE DUE:-\$ 181.93  
PAYMENT STATUS:UNPAID

SAVEPRINT



### 5.3.9.4 Activity Diagram

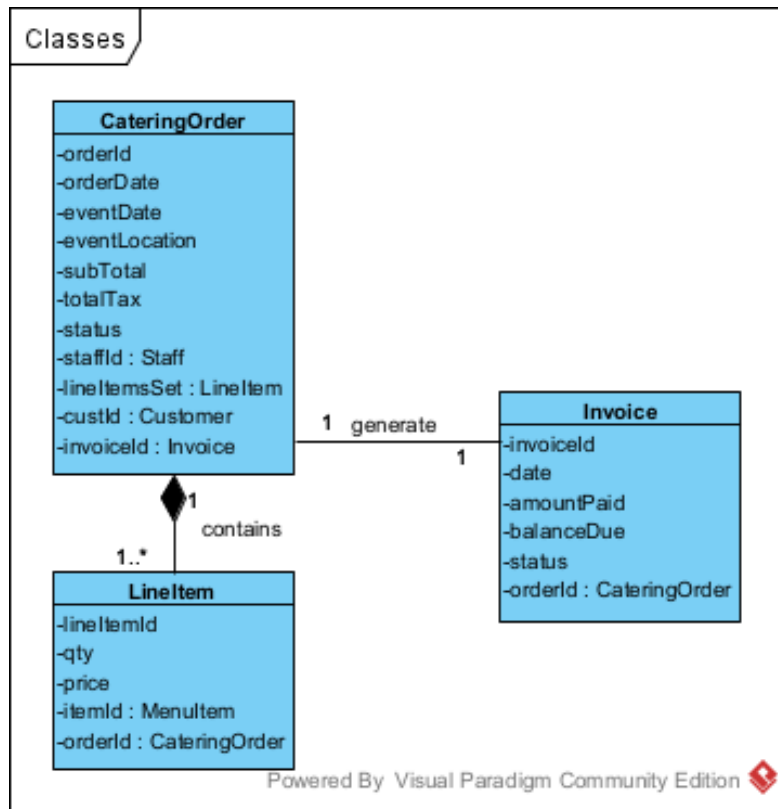


### 5.3.9.5 Use Case Description

|                   |                                                                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name     | Generate Catering Invoice                                                                                                                                                                                                                                                      |
| Triggering Event  | A confirmed catering order requires billing.                                                                                                                                                                                                                                   |
| Brief Description | The actor selects a confirmed catering order to generate a detailed catering invoice. The system displays the stored order details, including menu items, quantities, unit prices, subtotal, and tax, creates the invoice, initializes payment tracking fields, and stores it. |
| Actors            | Store Staff                                                                                                                                                                                                                                                                    |

|                      |                                                                                                                                                                                                                                                              |                                                                                               |                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Related Use Cases    | US-CO-01 Create Catering Order<br>US-CO-02 Update Catering Order<br>US-CO-07 Record Catering Payment                                                                                                                                                         |                                                                                               |                                                                                                                                                                                                                                                                                                                                                    |
| Preconditions        | The user is logged into the system.<br>The user has permission to generate catering invoices.<br>A catering order already exists in the system.<br>The catering order contains stored items, quantities, pricing, subtotal, tax, and total cost information. |                                                                                               |                                                                                                                                                                                                                                                                                                                                                    |
| Post Conditions      | A detailed catering invoice is generated successfully.<br>The invoice is stored securely in the system.<br>The invoice is linked to the related catering order.<br>The invoice includes payment tracking fields (Amount Paid, Balance Due, Payment Status).  |                                                                                               |                                                                                                                                                                                                                                                                                                                                                    |
| Flow of activities   | Actor                                                                                                                                                                                                                                                        |                                                                                               | System                                                                                                                                                                                                                                                                                                                                             |
|                      | 1.                                                                                                                                                                                                                                                           | Selects a catering order that requires billing.<br>Chooses the option to generate an invoice. | Validates that the order is confirmed and eligible for invoicing.<br>Retrieves the stored menu items, quantities, unit prices, subtotals, and total cost from the catering order.<br>Generate an invoice id.<br>Initializes payment fields (Amount Paid = 0, Balance Due = Total Amount, Status = Unpaid).<br>Display the details to double check. |
|                      | 2.                                                                                                                                                                                                                                                           | Reviews the displayed order information.<br>Save the invoice.                                 | Save the invoice.                                                                                                                                                                                                                                                                                                                                  |
| Exception Conditions | <ul style="list-style-type: none"> <li>• The actor cancels the process.</li> <li>• The system fails to generate or save the invoice due to a system error.</li> </ul>                                                                                        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                    |

### 5.3.9.6 Class Diagram



### 5.3.10 US-CO-07: Record Catering Payment

| Use Case ID | Use Case                | Use Case Description                                                                      | Trigger                                          | Actor       | Brief Description                                                                                                                                                                                     |
|-------------|-------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-CO-07    | Record Catering Payment | This use case allows Store Staff to manually record payments made for a catering invoice. | A customer makes a payment (deposit or balance). | Store Staff | The system retrieves the invoice, allows the staff to enter payment details, validates required fields, records the payment, updates the balance, and adjusts the invoice payment status accordingly. |

#### 5.3.10.1 User Story

As a Store Staff,

I want to record catering payments,

so that invoice balances are updated accurately and payment status reflects the actual transaction progress.

#### 5.3.10.2 Acceptance Criteria

1. Given catering invoices exist in the system, When the Store Staff accesses invoice records, Then the system displays available invoices for selection.
2. Given an invoice is selected, When the Store Staff enters payment amount, payment date, and payment method, Then the system records the payment successfully.
3. Given an invoice is selected, When the Store Staff enters missing or invalid payment information, Then the system displays an error and does not save the payment.
4. Given an invoice is selected, When the Store Staff records a payment, Then the system links the payment to the selected invoice.
5. Given a payment is successfully recorded, When the total payments are recalculated, Then the system updates the invoice payment status as Unpaid, Partially Paid, or Fully Paid and updates the remaining balance.

#### 5.3.10.3 UI Design (Mockup)

# Catering Order

| Order ID      | Customer     | Event Date | Order Status | Total Amount | Details                 | Invoice                  | Payment Status |
|---------------|--------------|------------|--------------|--------------|-------------------------|--------------------------|----------------|
| CO-2026-00125 | John Smith   | 2026-04-20 | Confirmed    | \$ 181.93    | <a href="#">DETAILS</a> | <a href="#">GENERATE</a> | Unpaid         |
| CO-2026-00126 | Alen Hunter  | 2026-03-31 | Delivered    | \$ 257.66    | <a href="#">DETAILS</a> | <a href="#">VIEW</a>     | Partially Paid |
| CO-2026-00127 | Jamie Oliver | 2026-03-27 | Delivered    | \$ 95.64     | <a href="#">DETAILS</a> | <a href="#">VIEW</a>     | Fully Paid     |
| CO-2026-00127 | Jamie Oliver | 2026-03-27 | Cancelled    | \$ 95.64     | <a href="#">DETAILS</a> |                          |                |

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# Invoice

INVOICE ID:CO-2026-00125-01

ORDER ID:CO-2026-00125  
CUSTOMER NAME:JOHN SMITH

EVENT DATE:2026-04-20  
ORDER STATUS:CONFIRMED

SELECTED ITEMS

| Item Name      | Qty | Unit Price | Subtotal |
|----------------|-----|------------|----------|
| Sandwich Tray  | 2   | \$45.00    | \$90.00  |
| Fruit Platter  | 1   | \$35.00    | \$35.00  |
| Drinks Package | 3   | \$12.00    | \$36.00  |

SUBTOTAL:

TAX (13%)

TOTAL AMOUNT:

AMOUNT PAID:

BALANCE DUE:

PAYMENT STATUS:

\$ 161.00

\$ 20.93

\$ 181.93

\$ 30.00

-\$ 151.93

PARTIALLY PAID

SAVE

PRINT

Invoice

INVOICE ID:CO-2026-00125-01

RECORD PAYMENT

|                         |                     |                             |
|-------------------------|---------------------|-----------------------------|
| PAYMENT DATE            | PAYMENT AMOUNT      | PAYMENT METHOD              |
| <div>2026 / 4 / 3</div> | <div>\$ 50.00</div> | <div>Cash</div> <div></div> |

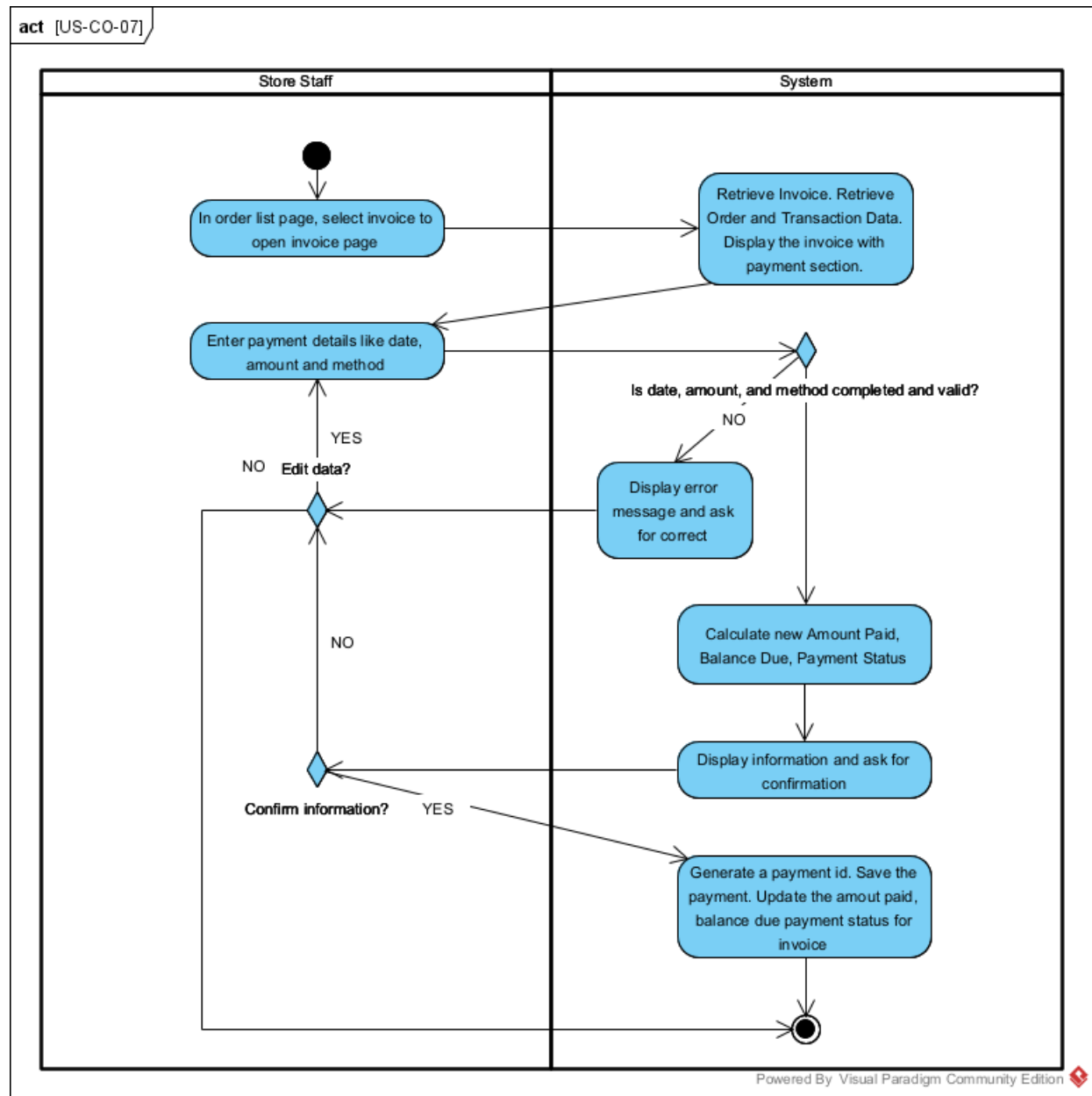
RECORD PAYMENT

PAYMENT HISTORY

- PAYMENT ID:CO-2026-00125-01-01    2026 / 3 / 20    \$ 30.00    CASH



### 5.3.10.4 Activity Diagram



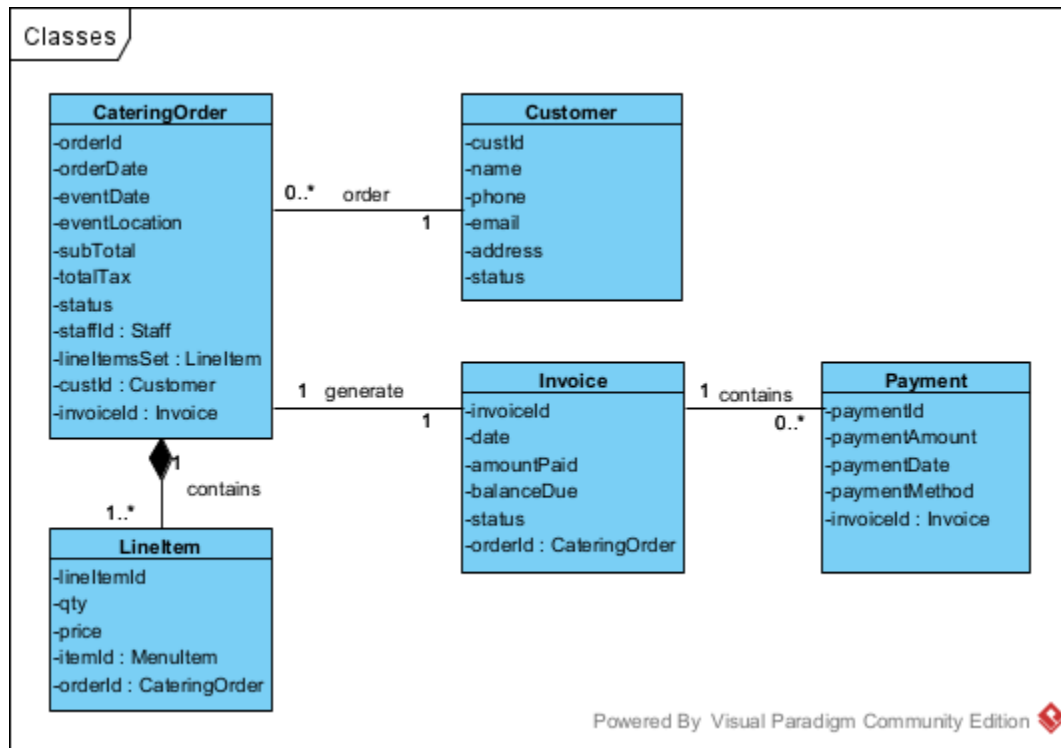
### 5.3.10.5 Use Case Description

|                   |                                                                                                                                                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name     | Record Catering Payment                                                                                                                                                                                                                                              |
| Triggering Event  | A customer makes a payment for a catering invoice.                                                                                                                                                                                                                   |
| Brief Description | The Store Staff selects an existing invoice and manually enters payment details. The system validates the input, associates the payment with the invoice, updates the total paid amount and remaining balance, and automatically adjusts the invoice payment status. |

|                      |                                                                                                                                                                                                                              |                                          |                                                                                                                                                                                       |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors               | Store Staff                                                                                                                                                                                                                  |                                          |                                                                                                                                                                                       |
| Related Use Cases    | US-CO-06 Generate Catering Invoice<br>US-CO-01 Create Catering Order                                                                                                                                                         |                                          |                                                                                                                                                                                       |
| Preconditions        | The user is logged into the system.<br>The user has permission to record payments.<br>A valid catering invoice exists in the system.                                                                                         |                                          |                                                                                                                                                                                       |
| Post Conditions      | The payment is recorded and stored in the system.<br>The payment is linked to the corresponding invoice.<br>The invoice balance is updated.<br>The invoice payment status is updated (Unpaid / Partially Paid / Fully Paid). |                                          |                                                                                                                                                                                       |
| Flow of activities   | Actor                                                                                                                                                                                                                        |                                          | System                                                                                                                                                                                |
|                      | 1.                                                                                                                                                                                                                           | Requests access to invoice records.      | Displays available catering invoices.                                                                                                                                                 |
|                      | 2.                                                                                                                                                                                                                           | Selects an invoice.                      | Displays invoice details.                                                                                                                                                             |
|                      | 3.                                                                                                                                                                                                                           | Enters payment amount, date, and method. | Validates payment information.<br>Calculate new Amount Paid, Balance Due, Payment Status.<br>Display information and ask for confirmation.                                            |
|                      | 4.                                                                                                                                                                                                                           | Confirms the payment.                    | Generate a payment id.<br>Records payment and links it to invoice.<br>Updates amount paid, balance due, and payment status.<br>Displays confirmation of successful payment recording. |
| Exception Conditions | <ul style="list-style-type: none"> <li>• The actor cancels the process.</li> <li>• The system fails to record the payment due to a system error.</li> </ul>                                                                  |                                          |                                                                                                                                                                                       |



### 5.3.10.6 Class Diagram



### 5.3.11 US-PM-01: Add New Menu Item

| Use Case ID | Use Case          | Use Case Description                                                                                            | Trigger                                                               | Actor          | Brief Description                                                                                         |
|-------------|-------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| US-PM-01    | Add New Menu Item | As a Business Owner, I want to add a new dessert or catering item to the system so that customers can order it. | The owner decides to introduce a new product or dish to the business. | Business Owner | The Business Owner enters the new item's details (name, description, price) and saves it to the database. |

#### 5.3.11.1 User Story

As a Business Owner.

I want to add new products and menu items to the system.

So that I can expand my catering and in-store offerings for customers.

#### 5.3.11.2 Acceptance Criteria

1. Given the Business Owner enters the new item details, then the system must validate that the item name, description, and price are not empty.
2. Given the Business Owner inputs the price, then the system must verify it is a valid positive number.
3. Given an existing menu item name is entered, then the system displays an error message to prevent duplicate entries.
4. Given all input data is valid and submitted, then the system must instantiate a new MenuItem object to store the record.
5. Given the transaction is complete, then the system saves the item to the database, displays a success message, and makes the item immediately available for new catering orders.

### 5.3.11.3 UI Design (Mockup)

Chocolatetown Business Management System

Back

Profile

Add New Menu Item

1. Item Details

Item Name:

Category:

Description:

Event Date: 2026-03-30 | Time: 14:00

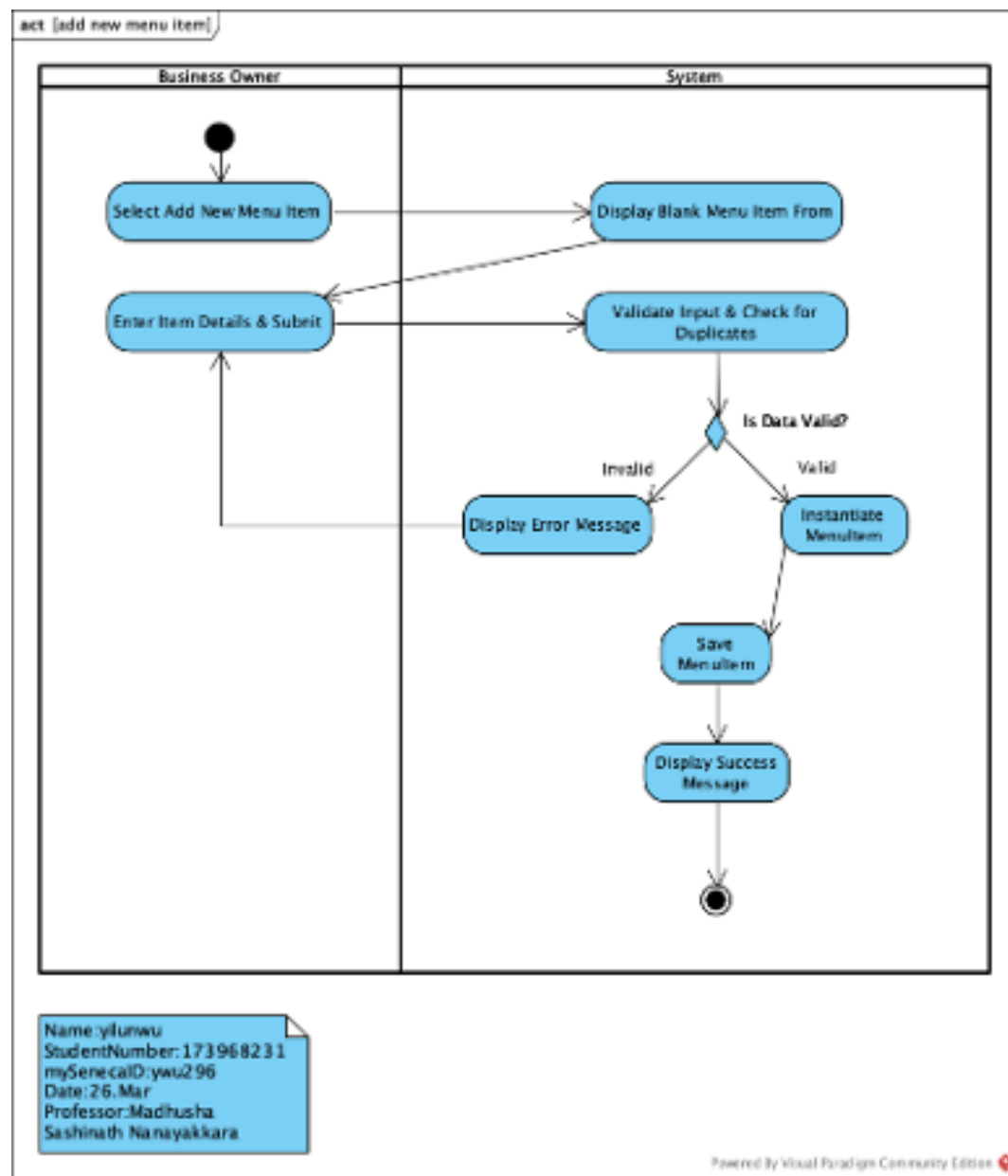
2. Pricing

Price:

CANCEL

CREATE ORDER

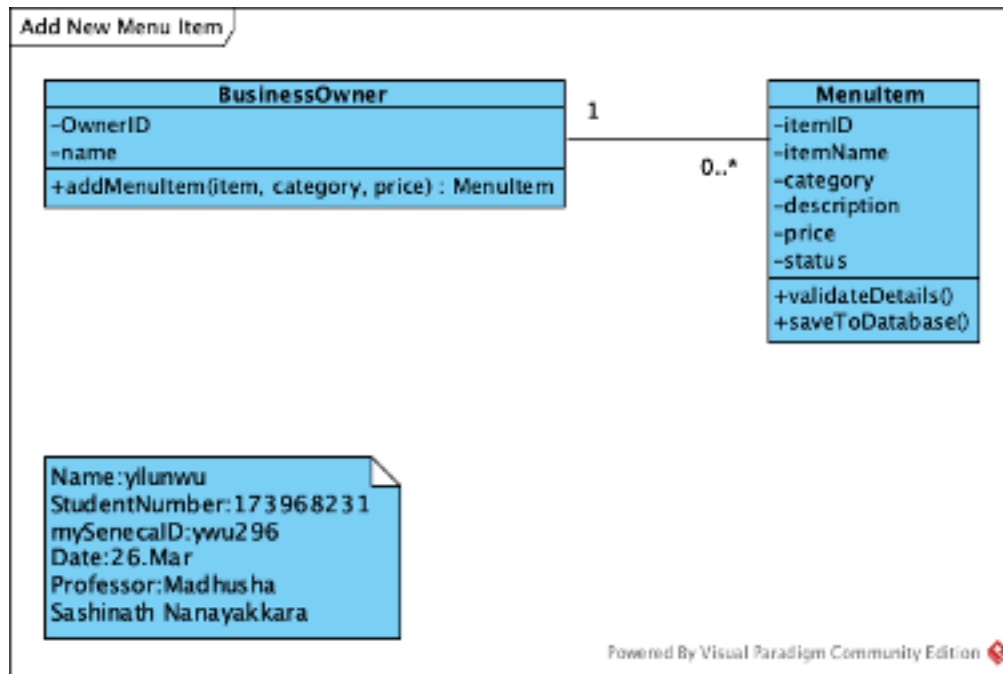
### 5.3.11.4 Activity Diagram



### 5.3.11.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                  |                                                                                                                                                   |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name        | Add New Menu Item                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                  |                                                                                                                                                   |
| Triggering Event     | The Business Owner decides to introduce a new dessert, snack, or catering product to the business.                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                  |                                                                                                                                                   |
| Brief Description    | Allows the Business Owner to enter product details (name, description, price) and save the new item to the system so it can be selected for future catering orders.                                                                                                                                                                                                                                                                                                                          |                                                                                                                                  |                                                                                                                                                   |
| Actors               | Business Owner                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                  |                                                                                                                                                   |
| Related Use Cases    | US-CO-01: Create Catering Order (Menu items created here are selected in US-CO-01)                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                  |                                                                                                                                                   |
| Preconditions        | The Business Owner has successfully logged into the system with administrative privileges.                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                  |                                                                                                                                                   |
| Post Conditions      | A new MenuItem object is instantiated and permanently saved to the database. The new item is immediately available for selection in the system.                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                  |                                                                                                                                                   |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                  | System                                                                                                                                            |
|                      | 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Selects the option to add a new menu item from the product management dashboard.                                                 | Displays a blank "New Menu Item" entry form.                                                                                                      |
|                      | 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Enters the item details, including Item Name, Description, Category (e.g., Fondue, Ice Cream), and Price, then submits the form. | Validates the input data to ensure no mandatory fields are empty, the price is a valid positive number, and the item name does not already exist. |
|                      | 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                  | Instantiates a new MenuItem object with the provided details. Saves the record to the database and displays a "Success" confirmation message.     |
| Exception Conditions | <ul style="list-style-type: none"> <li>● <b>Missing or Invalid Data:</b> If mandatory fields are empty or the price format is invalid (e.g., negative number or text), the system displays an error message and highlights the required fields for correction.</li> <li>● <b>Duplicate Item Name:</b> If the system detects that the entered item name already exists in the database, it alerts the Business Owner and prevents the record from being saved to avoid duplicates.</li> </ul> |                                                                                                                                  |                                                                                                                                                   |

### 5.3.11.6 Class Diagram



### 5.3.12 US-PM-02: Update Menu Item

| Use Case ID | Use Case         | Use Case Description                                                                                                                                     | Trigger                                                                      | Actor          | Brief Description                                                                                                                                                                             |
|-------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-PM-02    | Update Menu Item | As a Business Owner, I want to update the details of an existing menu item so that the system reflects the most current pricing and product information. | Pricing, description, or name of an existing menu item needs to be modified. | Business Owner | The Business Owner selects an existing menu item, updates its details (such as price or description), and saves the changes. The system ensures previously generated invoices are unaffected. |

#### 5.3.12.1 User Story

As a Business Owner

I want to update the pricing and descriptions of existing menu items.

So that my catering menu reflects current costs and offerings, while ensuring past financial records remain accurate.

#### 5.3.12.2 Acceptance Criteria

1. Given the Business Owner selects an active menu item to edit, then the system must retrieve and display its current details (name, description, category, and price).
2. Given the Business Owner updates the item price, then the system must validate that the new price is a valid positive number.
3. Given the Business Owner modifies the menu item details, then the system must validate that mandatory fields (such as item name and price) are not left empty.
4. Given the item's price is successfully modified, then the system must ensure this price change does not alter the total cost of any previously generated catering invoices (Based on BR16).
5. Given all updated information is valid and submitted, then the system saves the modifications to the database and displays a "Success" confirmation message.

### 5.3.12.3 UI Design (Mockup)

Chocolatetown Business Management System

Back

Profile

Update Menu Item

1. Select Item to Edit

Premium Chocolate Fondue Set

Status: Active

2. Item Details

Category: Desserts

Description: Our signature rich chocolate

marshmallows.

3. Pricing

Price (\$): 1500

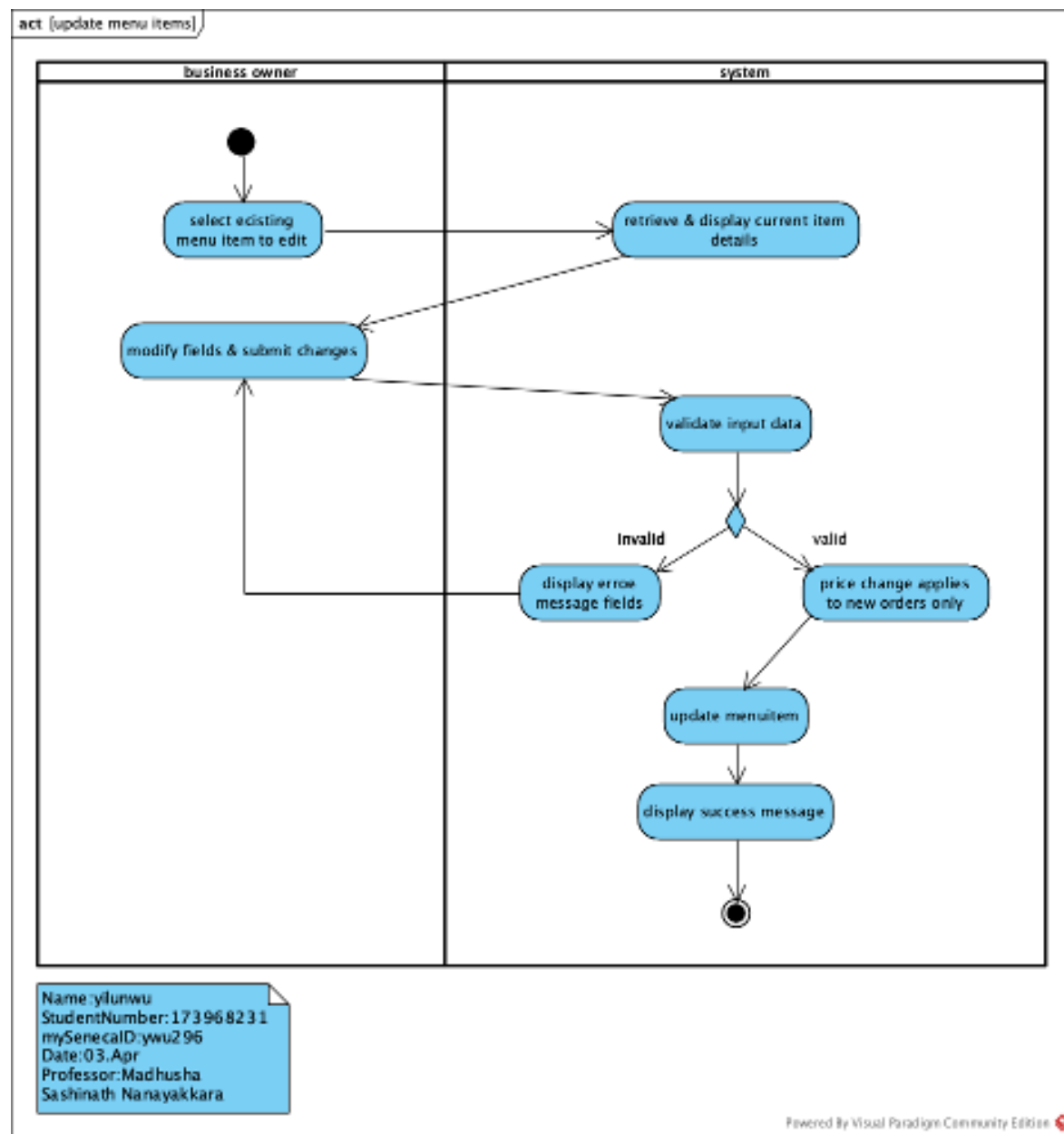
Note: Pricing modifications will only apply to new catering orders. Previously generated invoices will remain unaffected.

CANCEL

SAVE CHANGES



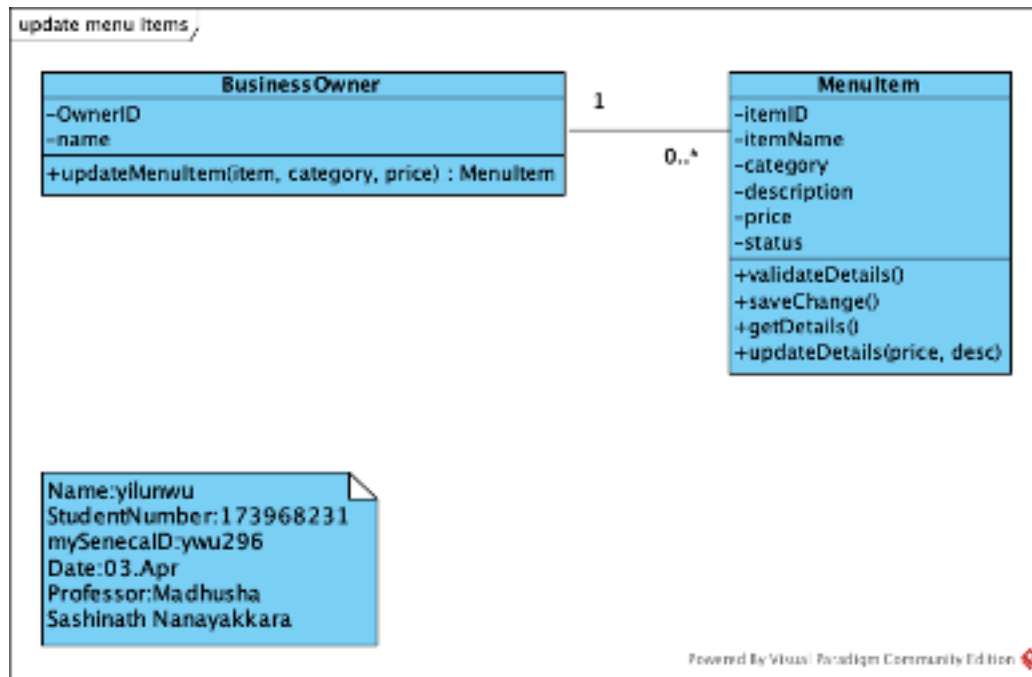
### 5.3.12.4 Activity Diagram



### 5.3.12.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                 |                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Use Case Name        | Update Menu Item                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                 |                                                                                                                          |
| Triggering Event     | Pricing or description needs to be modified.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                 |                                                                                                                          |
| Brief Description    | The item's details are updated while maintaining its history in the system. The Business Owner modifies the fields, and the system saves the updated information while ensuring historical invoice data remains unaffected.                                                                                                                                                                                                                                                                  |                                                                                                 |                                                                                                                          |
| Actors               | Business Owner                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                 |                                                                                                                          |
| Related Use Cases    | US-PM-01: Add New Menu Item (Pre-requisite)<br>US-CO-01: Create Catering Order (Uses the updated pricing for future orders)<br>US-CO-06: Generate Catering Invoice (Must not be affected by price changes )                                                                                                                                                                                                                                                                                  |                                                                                                 |                                                                                                                          |
| Preconditions        | The Business Owner has logged into the system with administrative privileges.<br>The menu item already exists in the system database.                                                                                                                                                                                                                                                                                                                                                        |                                                                                                 |                                                                                                                          |
| Post Conditions      | The existing MenuItem object is updated in the database. Menu pricing modifications shall not affect previously generated catering invoices.                                                                                                                                                                                                                                                                                                                                                 |                                                                                                 |                                                                                                                          |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                 | System                                                                                                                   |
|                      | 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Requests to view the menu item list and selects a specific item to edit..                       | Retrieves the MenuItem object and displays its current details (Name, Category, Description, Price) in an editable form. |
|                      | 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Updates the desired fields (e.g., changes the price to reflect new costs) and submits the form. | Validates the input data to ensure no mandatory fields are empty and the price is a valid positive number.               |
|                      | 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                 | Updates the MenuItem object in the database with the new details.                                                        |
|                      | 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                 | Displays a "Success" confirmation message and ensures the new pricing is applied only to future orders.                  |
| Exception Conditions | <ul style="list-style-type: none"> <li>● <b>Missing or Invalid Data:</b> If mandatory fields are cleared or the updated price format is invalid (e.g., negative number or text), the system displays an error message, highlights the fields, and prevents the update.</li> <li>● <b>Duplicate Item Name:</b> If the Business Owner changes the item name to one that already exists in the system, the system alerts the user and prevents the update to avoid naming conflicts.</li> </ul> |                                                                                                 |                                                                                                                          |

### 5.3.12.6 Class Diagram



### 5.3.13 US-SM-01: Add New Supplier

| Use Case ID | Use Case         | Use Case Description                                                                                                                                 | Trigger                                           | Actor                | Brief Description                                                                                                                                                                                                                                 |
|-------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-SM-01    | Add New Supplier | This use case describes how the Supplier Coordinator adds a new supplier record into the system when Chocolatetown starts working with a new vendor. | A new partnership is established with a supplier. | Supplier Coordinator | The Supplier Coordinator enters supplier information such as supplier name, contact person, phone number, address, supplied product category, and payment details. The system validates the input and saves a new supplier profile in the system. |

#### 5.3.13.1 User Story

As a staff member,

I want to add a new supplier to the system,

so that I can manage supplier information and support business operations efficiently.

#### 5.3.13.2 Acceptance Criteria

1. Given the Supplier Coordinator opens the supplier transaction form, when the form is displayed, then the system shall allow the user to enter transaction details.
2. Given the Supplier Coordinator selects an existing active supplier and enters valid transaction information, when the form is submitted, then the system shall create a new supplier transaction record successfully.
3. Given required fields such as supplier, transaction date, or amount are missing, when the user submits the form, then the system shall prevent submission and display an error message.
4. Given the selected supplier is inactive, when the user tries to submit the transaction, then the system shall prevent the transaction from being saved and notify the user.
5. Given the supplier transaction is saved successfully, when the transaction is complete, then the system shall store the record for future invoice, payment, and audit tracking.

### 5.3.13.3 UI Design (Mockup)

**Add New Supplier**

Supplier Name:

Contact Person:

Phone Number:

Email:

Address:

Product Category:

Payment Details:

Status:

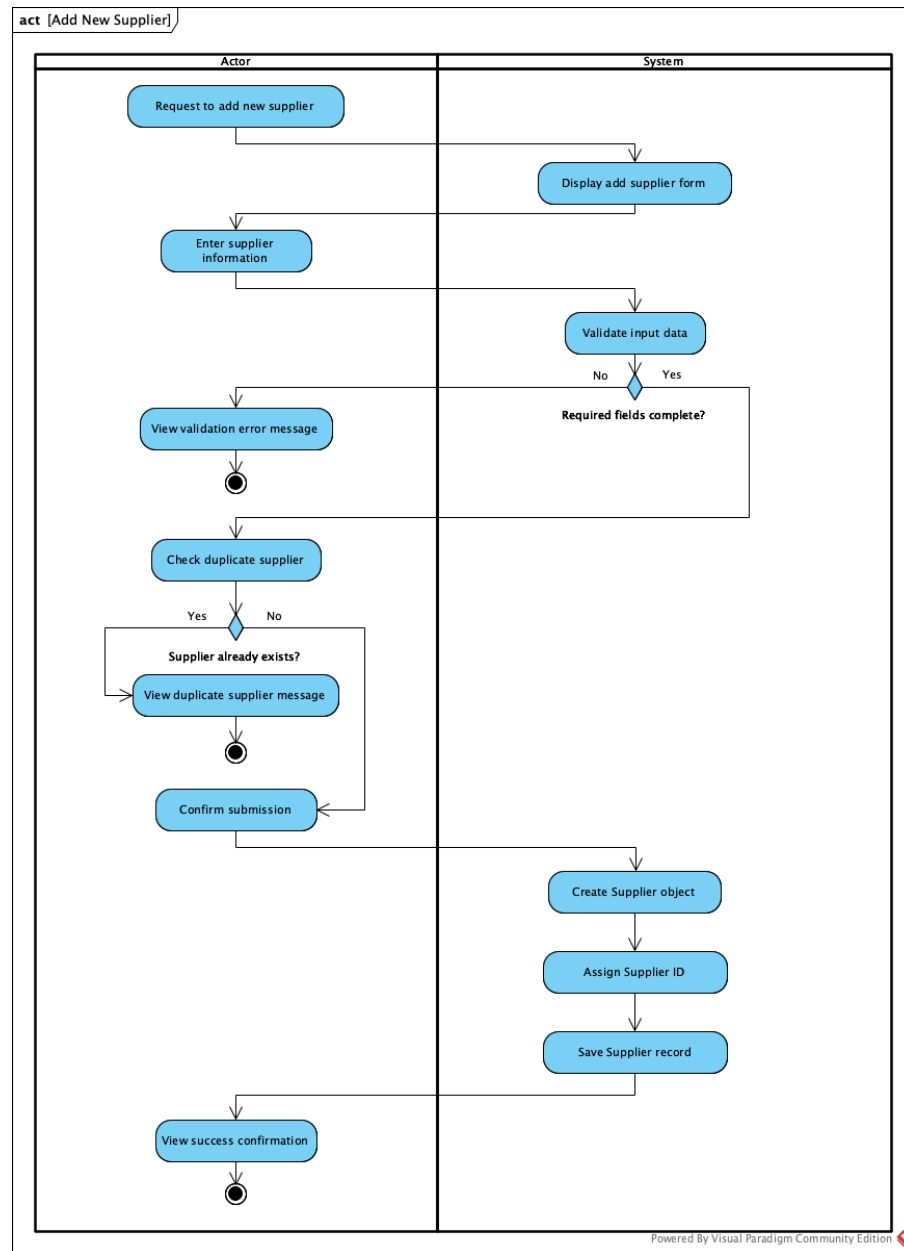
Active

▼

Save Supplier

Cancel

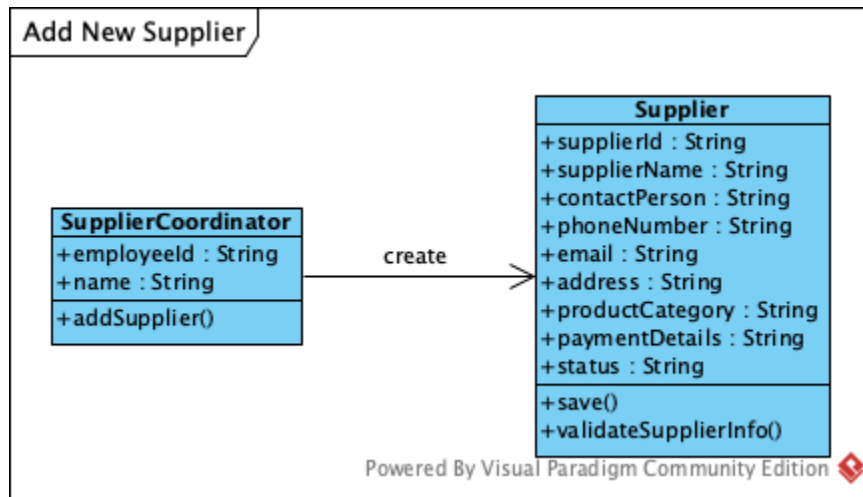
### 5.3.13.4 Activity Diagram



### 5.3.13.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                     |                                                                                                                                         |                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Use Case Name        | Add New Supplier                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |                                                                             |
| Triggering Event     | A new supplier starts working with the business.                                                                                                                                                                                                                                                                    |                                                                                                                                         |                                                                             |
| Brief Description    | Allows the Supplier Coordinator to enter supplier details and save a new supplier record into the system for future transaction and payment tracking.                                                                                                                                                               |                                                                                                                                         |                                                                             |
| Actors               | Supplier Coordinator                                                                                                                                                                                                                                                                                                |                                                                                                                                         |                                                                             |
| Related Use Cases    | US-SM-04: Record Supplier Transaction<br>US-SM-05: Upload Supplier Invoice<br>US-SM-06: Record Supplier Payment                                                                                                                                                                                                     |                                                                                                                                         |                                                                             |
| Preconditions        | The Supplier Coordinator has logged into the system.<br>The Supplier Coordinator has permission to manage supplier records.                                                                                                                                                                                         |                                                                                                                                         |                                                                             |
| Post Conditions      | A new Supplier object is instantiated and stored in the database with active status.                                                                                                                                                                                                                                |                                                                                                                                         |                                                                             |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                               |                                                                                                                                         | System                                                                      |
|                      | 1.                                                                                                                                                                                                                                                                                                                  | Selects the option to add a new supplier.                                                                                               | Displays the add supplier form.                                             |
|                      | 2.                                                                                                                                                                                                                                                                                                                  | Enters supplier information such as supplier name, contact person, phone number, email, address, product category, and payment details. | Validates the entered data and checks whether required fields are complete. |
|                      | 3.                                                                                                                                                                                                                                                                                                                  | Submits the form.                                                                                                                       | Checks for duplicate supplier records.                                      |
|                      | 4.                                                                                                                                                                                                                                                                                                                  |                                                                                                                                         | Generates a unique supplier ID.                                             |
|                      | 5.                                                                                                                                                                                                                                                                                                                  |                                                                                                                                         | Instantiates a new Supplier object and saves it to the database.            |
|                      | 6.                                                                                                                                                                                                                                                                                                                  |                                                                                                                                         | Displays a success confirmation message.                                    |
| Exception Conditions | <ul style="list-style-type: none"> <li>Missing Mandatory Data: If required fields are empty, the system displays an error message and prevents submission.</li> <li>Duplicate Supplier: If the supplier already exists in the system, the system shows a duplicate warning and does not save the record.</li> </ul> |                                                                                                                                         |                                                                             |

### 5.3.13.6 Class Diagram





### 5.3.14 US-SM-04: Record Supplier Transaction

| Use Case ID | Use Case                    | Use Case Description                                                                                                                                                                                    | Trigger                                                                   | Actor                | Brief Description                                                                                                                                                                                                                                                                  |
|-------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-SM-04    | Record Supplier Transaction | This use case describes how the Supplier Coordinator records a new supplier transaction in the system when Chocolatetown purchases ingredients, packaging materials, or other supplies from a supplier. | A new supplier transaction occurs and needs to be recorded in the system. | Supplier Coordinator | The Supplier Coordinator selects a supplier, enters transaction details such as transaction date, amount, supplied items, and notes, and submits the transaction. The system validates the input and saves the new supplier transaction record for tracking and auditing purposes. |

#### 5.3.14.1 User Story

As a Supplier Coordinator,  
I want to record a supplier transaction in the system,  
so that I can keep accurate purchasing records and track supplier-related business activities in an organized way.

#### 5.3.14.2 Acceptance Criteria

1. Given the Supplier Coordinator opens the supplier transaction form, then the system shall allow the user to enter transaction details.
2. Given the Supplier Coordinator selects an existing active supplier and enters valid transaction information, then the system shall create a new supplier transaction record successfully.
3. Given required fields such as supplier, transaction date, or amount are missing, then the system shall prevent submission and display an error message.
4. Given the selected supplier is inactive, then the system shall prevent the transaction from being saved and notify the user.
5. Given the supplier transaction is saved successfully, then the system shall store the record for future invoice, payment, and audit tracking.

### 5.3.14.3 UI Design (Mockup)

Record Supplier Transaction

Supplier:

— Select Supplier —

▼

Transaction Date:

yyyy/mm/dd

📅

Transaction Amount:

0.00

Items / Materials:

e.g. cocoa powder, foil wrappers

Notes:

Optional

Status:

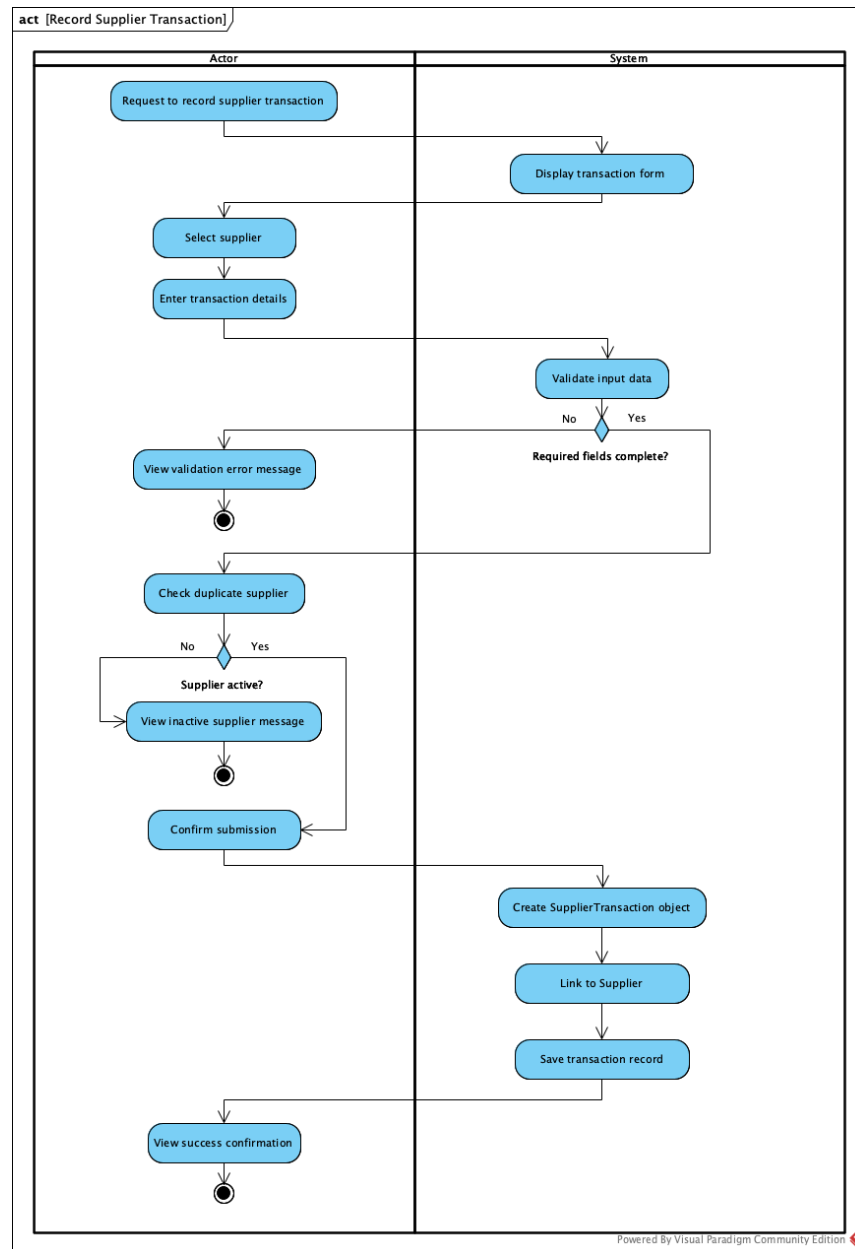
Recorded

▼

Save Transaction

Cancel

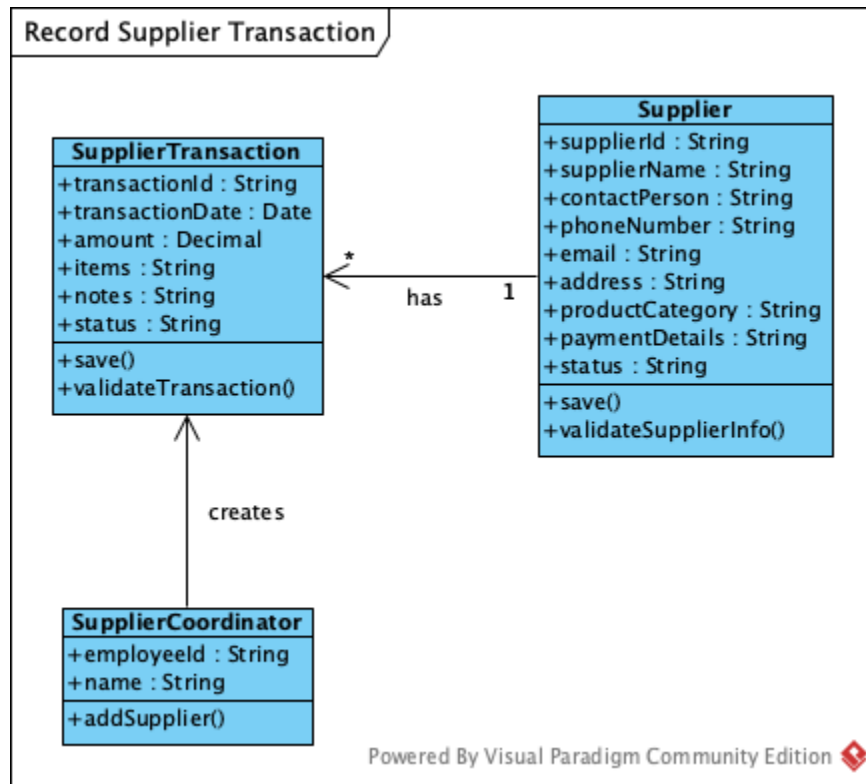
### 5.3.14.4 Activity Diagram



### 5.3.14.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         |                                                                                 |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Use Case Name        | Record Supplier Transaction                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                 |
| Triggering Event     | A new purchase or supply transaction with a supplier needs to be recorded.                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                         |                                                                                 |
| Brief Description    | Allows the Supplier Coordinator to select a supplier, enter transaction details, and save the supplier transaction in the system for future invoice, payment, and audit tracking.                                                                                                                                                                                                                                                                                                                      |                                                                                         |                                                                                 |
| Actors               | Supplier Coordinator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                         |                                                                                 |
| Related Use Cases    | US-SM-01: Add New Supplier<br>US-SM-05: Upload Supplier Invoice<br>US-SM-06: Record Supplier Payment                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                         |                                                                                 |
| Preconditions        | The Supplier Coordinator has logged into the system.<br>The selected supplier already exists in the system.<br>The selected supplier is active.<br>The Supplier Coordinator has permission to manage supplier transactions.                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                 |
| Post Conditions      | A new <b>SupplierTransaction</b> object is instantiated and saved in the database, linked to the selected supplier.                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                         |                                                                                 |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                         | System                                                                          |
|                      | 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Selects the option to record a supplier transaction.                                    | Displays the supplier transaction entry form.                                   |
|                      | 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Selects an existing supplier from the list.                                             | Retrieves supplier information and prepares the transaction form.               |
|                      | 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enters transaction details such as transaction date, amount, supplied items, and notes. | Validates the entered data and checks whether all required fields are complete. |
|                      | 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Submits the form.                                                                       | Checks whether the selected supplier is active.                                 |
|                      | 5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                         | Instantiates a new SupplierTransaction object linked to the supplier.           |
|                      | 6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                         | Saves the transaction record to the database.                                   |
|                      | 7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                         | Displays a success confirmation message.                                        |
| Exception Conditions | <ul style="list-style-type: none"> <li>Missing Mandatory Data: If supplier, transaction date, or amount is missing, the system displays an error message and prevents submission.</li> <li>Inactive Supplier: If the selected supplier is inactive, the system prevents the transaction from being saved and displays a warning message.</li> <li>Invalid Amount: If the amount is empty, zero, or not a valid number, the system displays an error and asks the user to correct the value.</li> </ul> |                                                                                         |                                                                                 |

### 5.3.14.6 Class Diagram



### 5.3.15 US-SM-05: Upload Supplier Invoice

| Use Case ID | Use Case                | Use Case Description                                                                                                                                  | Trigger                                                  | Actor                | Brief Description                                                                                                                                                                                                                                                                                    |
|-------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-SM-05    | Upload Supplier Invoice | This use case describes how the Supplier Coordinator uploads or records a supplier invoice related to an existing supplier transaction in the system. | A supplier invoice is received and needs to be recorded. | Supplier Coordinator | The Supplier Coordinator selects an existing supplier transaction, enters invoice information such as invoice number, invoice date, total amount, and attachment, and submits the invoice record. The system validates the input and stores the invoice linked to the selected supplier transaction. |

#### 5.3.15.1 User Story

As a Supplier Coordinator,  
I want to upload or record a supplier invoice,  
so that I can keep invoice records organized and linked to the correct supplier transaction.

#### 5.3.15.2 Acceptance Criteria

1. Given the Supplier Coordinator opens the supplier invoice form, then the system shall allow the user to enter invoice details and upload an invoice file.
2. Given the Supplier Coordinator selects an existing supplier transaction and enters valid invoice information, then the system shall create a new supplier invoice record successfully.
3. Given the selected supplier transaction does not exist, then the system shall prevent the invoice from being recorded and display an error message.
4. Given required fields such as invoice number, invoice date, or amount are missing, then the system shall prevent submission and display a validation error.
5. Given the supplier invoice is recorded successfully, then the system shall store the invoice and link it to the selected supplier transaction for future tracking and auditing.

### 5.3.15.3 UI Design (Mockup)

Record Supplier Transaction

Supplier:

— Select Supplier —

▼

Transaction Date:

yyyy/mm/dd

📅

Transaction Amount:

0.00

Items / Materials:

Notes:

Optional

Status:

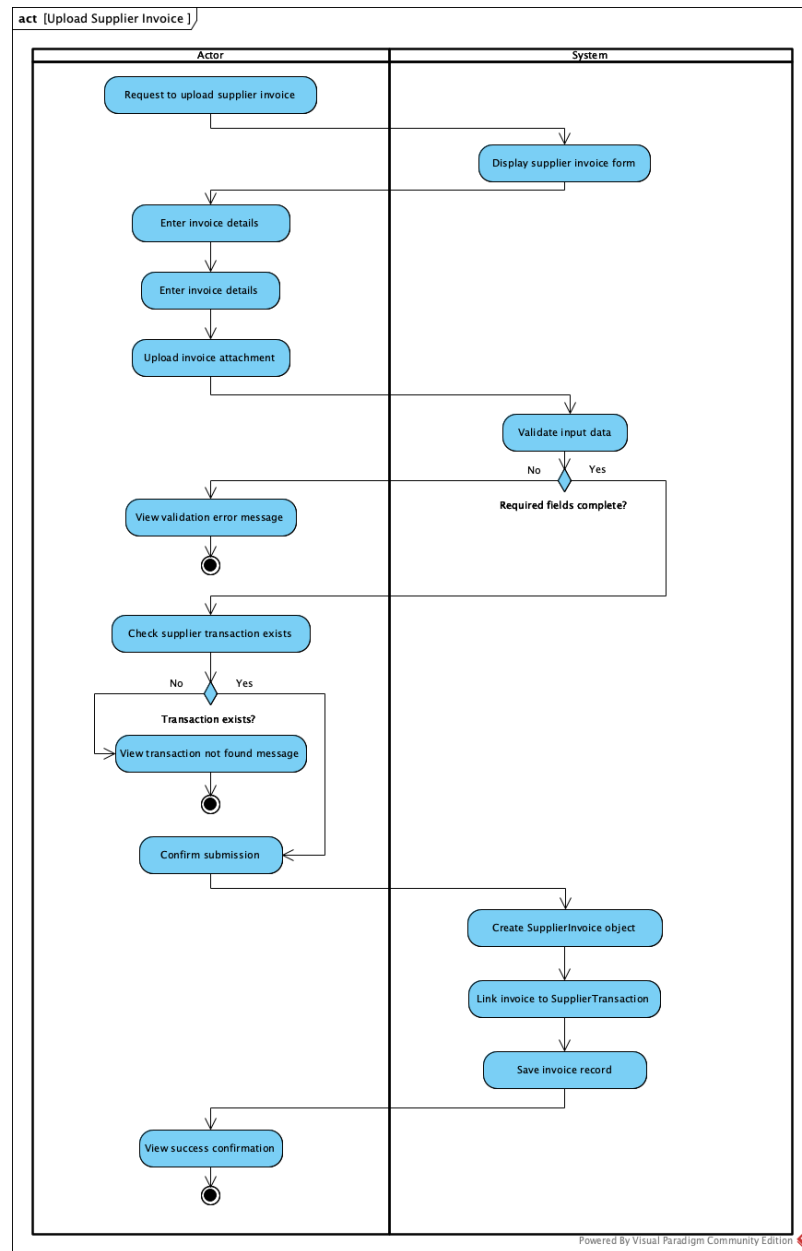
Recorded

▼

Save Transaction

Cancel

### 5.3.15.4 Activity Diagram



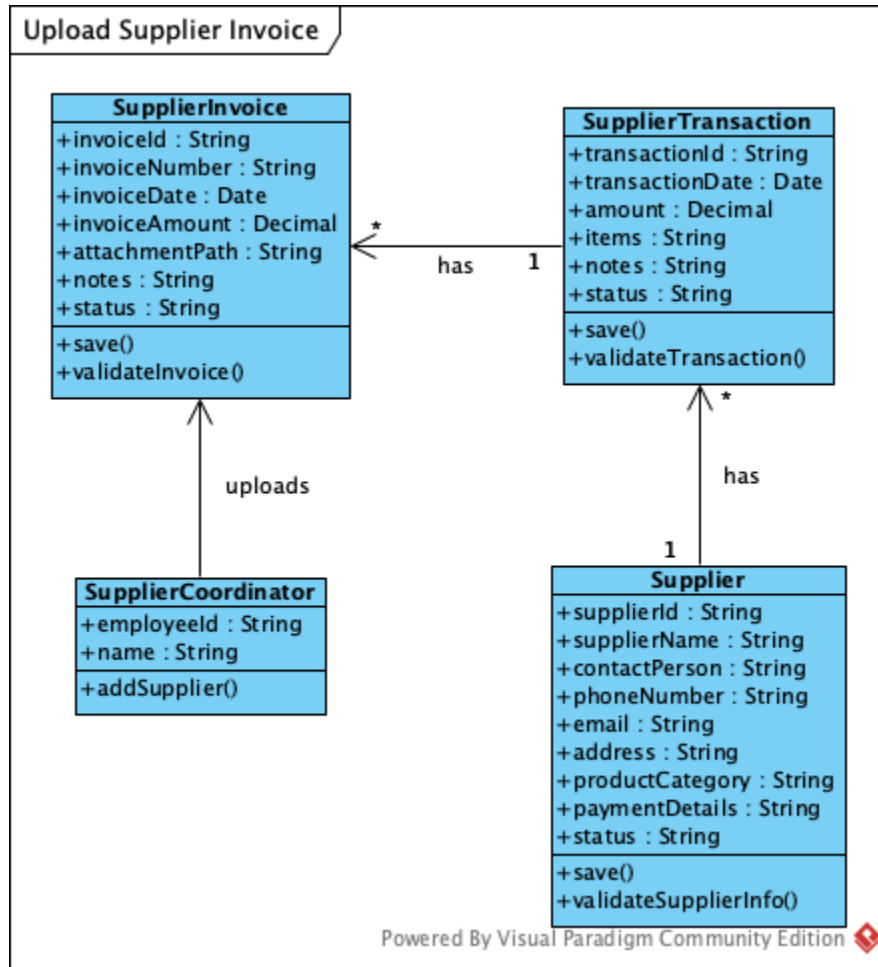


### 5.3.15.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                 |                                                                                        |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Use Case Name        | Upload Supplier Invoice                                                                                                                                                                                                                                                                                                                                                 |                                                                                 |                                                                                        |
| Triggering Event     | A supplier invoice is received and needs to be recorded in the system.                                                                                                                                                                                                                                                                                                  |                                                                                 |                                                                                        |
| Brief Description    | Allows the Supplier Coordinator to select an existing supplier transaction, enter invoice details, upload an invoice file if needed, and save the invoice record in the system.                                                                                                                                                                                         |                                                                                 |                                                                                        |
| Actors               | Supplier Coordinator                                                                                                                                                                                                                                                                                                                                                    |                                                                                 |                                                                                        |
| Related Use Cases    | <ul style="list-style-type: none"> <li>• US-SM-04: Record Supplier Transaction</li> <li>• US-SM-06: Record Supplier Payment</li> </ul>                                                                                                                                                                                                                                  |                                                                                 |                                                                                        |
| Preconditions        | <ul style="list-style-type: none"> <li>• The Supplier Coordinator has logged into the system.</li> <li>• The selected supplier transaction already exists in the system.</li> <li>• The Supplier Coordinator has permission to manage supplier invoices.</li> </ul>                                                                                                     |                                                                                 |                                                                                        |
| Post Conditions      | A new SupplierInvoice object is instantiated and saved in the database, linked to the selected SupplierTransaction.                                                                                                                                                                                                                                                     |                                                                                 |                                                                                        |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                                                                                   | System                                                                          |                                                                                        |
|                      | 1.                                                                                                                                                                                                                                                                                                                                                                      | Selects the option to upload a supplier invoice.                                | Displays the supplier invoice form.                                                    |
|                      | 2.                                                                                                                                                                                                                                                                                                                                                                      | Selects an existing supplier transaction.                                       | Retrieves transaction information and prepares the invoice form.                       |
|                      | 3.                                                                                                                                                                                                                                                                                                                                                                      | Enters invoice details such as invoice number, invoice date, amount, and notes. | Validates the entered data and checks whether required fields are complete.            |
|                      | 4.                                                                                                                                                                                                                                                                                                                                                                      | Uploads the invoice attachment and submits the form.                            | Checks whether the selected supplier transaction exists.                               |
|                      | 5.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                 | Instantiates a new SupplierInvoice object linked to the selected supplier transaction. |
|                      | 6.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                 | Saves the invoice record to the database.                                              |
|                      | 7.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                 | Displays a success confirmation message.                                               |
| Exception Conditions | <ul style="list-style-type: none"> <li>• Missing Mandatory Data: If invoice number, invoice date, or amount is missing, the system displays an error message and prevents submission.</li> <li>• Transaction Not Found: If the selected supplier transaction does not exist, the system prevents the invoice from being saved and displays an error message.</li> </ul> |                                                                                 |                                                                                        |

|  |                                                                                                                                                                                                     |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>Invalid Amount: If the invoice amount is empty, zero, or not a valid number, the system displays an error and asks the user to correct the value.</li> </ul> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.3.15.6 Class Diagram



### 5.3.16 US-SM-06: Record Supplier Payment

| Use Case ID | Use Case                | Use Case Description                                                                                                             | Trigger                                                   | Actor                | Brief Description                                                                                                                                                                                                                                                                           |
|-------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-SM-06    | Record Supplier Payment | This use case describes how the Supplier Coordinator records a payment made to a supplier for a supplier transaction or invoice. | A payment is made to a supplier and needs to be recorded. | Supplier Coordinator | The Supplier Coordinator selects the related supplier transaction or invoice, enters payment details such as payment amount, payment date, and payment method, and submits the payment record. The system validates the input and stores the payment linked to the correct supplier record. |

#### 5.3.16.1 User Story

As a Supplier Coordinator,

I want to record supplier payments in the system,

so that I can keep accurate payment records for suppliers and support financial tracking.

#### 5.3.16.2 Acceptance Criteria

1. Given the Supplier Coordinator opens the supplier payment form, then the system shall allow the user to enter payment details.
2. Given the Supplier Coordinator selects a valid supplier transaction or invoice and enters valid payment information, then the system shall create a new supplier payment record successfully.
3. Given required fields such as payment amount, payment date, or payment method are missing, then the system shall prevent submission and display an error message.
4. Given the related supplier transaction or invoice does not exist, then the system shall prevent the payment from being saved and notify the user.
5. Given the supplier payment is recorded successfully, then the system shall store the payment record and link it to the related supplier transaction or invoice for future financial review and auditing.

### 5.3.16.3 UI Design (Mockup)

### Upload Supplier Invoice

Supplier Transaction:

— Select Transaction —

▼

Invoice Number:

Invoice Date:

yyyy/mm/dd

Invoice Amount:

0.00

Attachment:

Choose File

No file chosen

Notes:

Optional

Status:

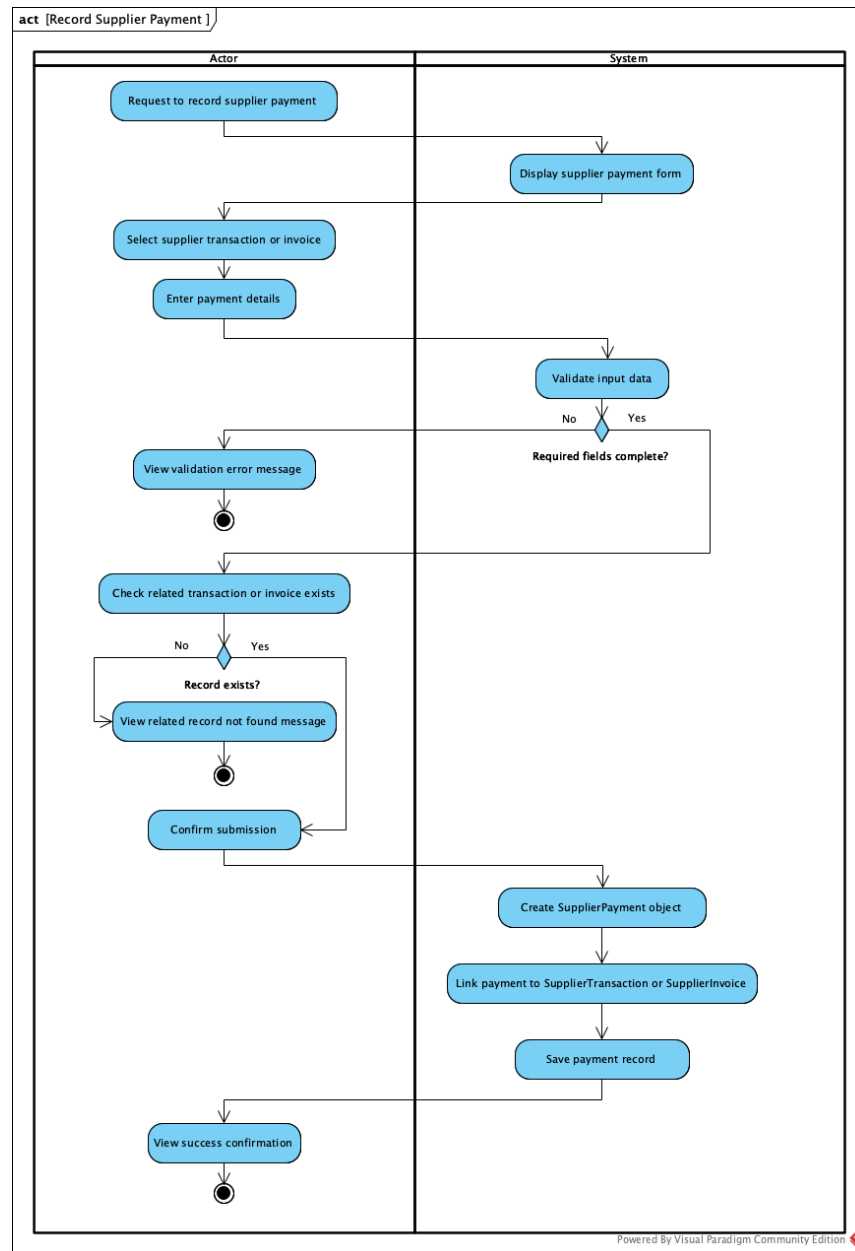
Recorded

▼

Save Invoice

Cancel

### 5.3.16.4 Activity Diagram

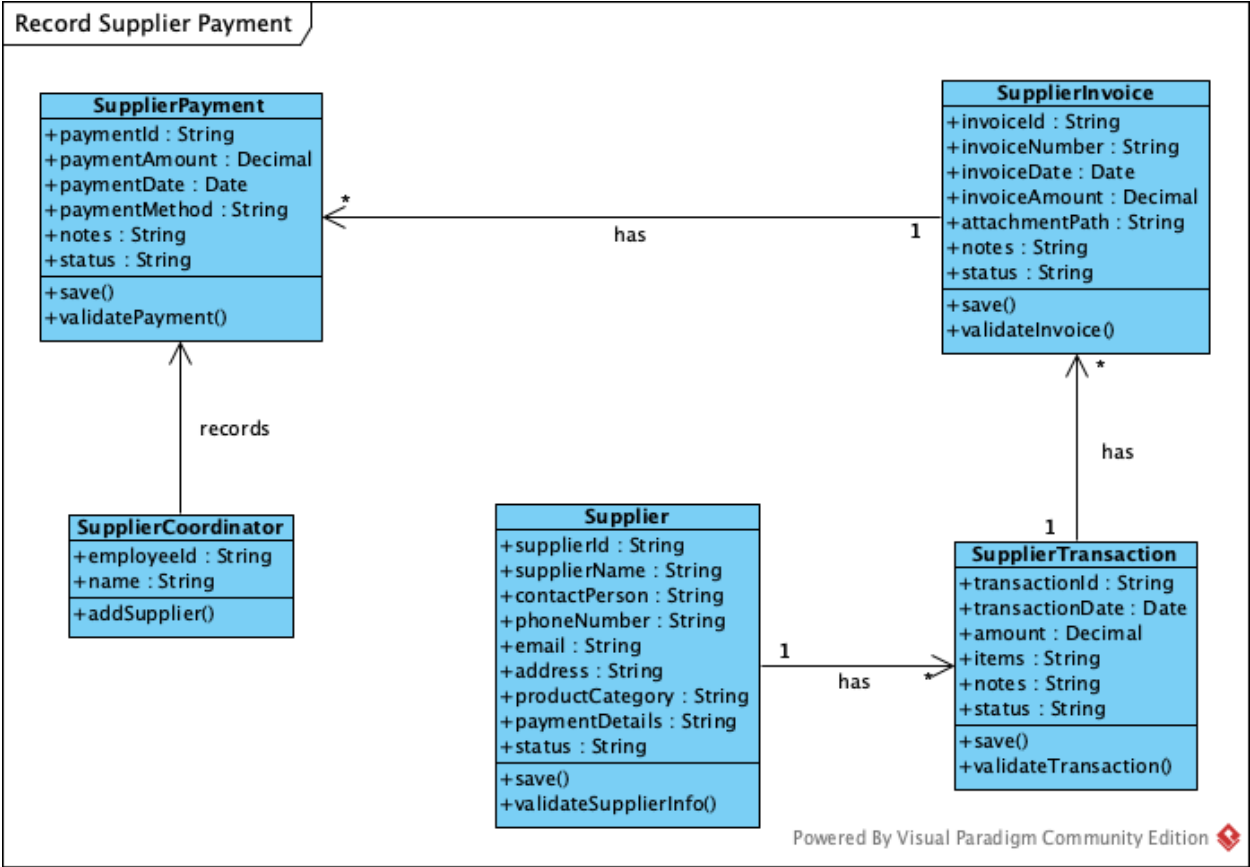


### 5.3.16.5 Use Case Description

|                      |                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                         |                                                                                  |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Use Case Name        | Record Supplier Payment                                                                                                                                                                                                                                                                                                                                                                   |                                                                                         |                                                                                  |
| Triggering Event     | A payment is made to a supplier and needs to be recorded in the system.                                                                                                                                                                                                                                                                                                                   |                                                                                         |                                                                                  |
| Brief Description    | Allows the Supplier Coordinator to select the related supplier invoice, enter payment details, and save the supplier payment record in the system.                                                                                                                                                                                                                                        |                                                                                         |                                                                                  |
| Actors               | Supplier Coordinator                                                                                                                                                                                                                                                                                                                                                                      |                                                                                         |                                                                                  |
| Related Use Cases    | <ul style="list-style-type: none"> <li>US-SM-04: Record Supplier Transaction</li> <li>US-SM-05: Upload Supplier Invoice</li> </ul>                                                                                                                                                                                                                                                        |                                                                                         |                                                                                  |
| Preconditions        | <ul style="list-style-type: none"> <li>The Supplier Coordinator has logged into the system.</li> <li>The related supplier transaction or supplier invoice already exists in the system.</li> <li>The Supplier Coordinator has permission to manage supplier payment records.</li> </ul>                                                                                                   |                                                                                         |                                                                                  |
| Post Conditions      | A new SupplierPayment object is instantiated and saved in the database, linked to the related SupplierTransaction or SupplierInvoice.                                                                                                                                                                                                                                                     |                                                                                         |                                                                                  |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                                                                                                     | System                                                                                  |                                                                                  |
|                      | 1.                                                                                                                                                                                                                                                                                                                                                                                        | Selects the option to record a supplier payment.                                        | Displays the supplier payment form.                                              |
|                      | 2.                                                                                                                                                                                                                                                                                                                                                                                        | Selects the related supplier transaction or supplier invoice.                           | Retrieves related record information and prepares the payment form.              |
|                      | 3.                                                                                                                                                                                                                                                                                                                                                                                        | Enters payment details such as payment amount, payment date, payment method, and notes. | Validates the entered data and checks whether required fields are complete.      |
|                      | 4.                                                                                                                                                                                                                                                                                                                                                                                        | Submits the form.                                                                       | Checks whether the related supplier transaction or invoice exists.               |
|                      | 5.                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         | Instantiates a new SupplierPayment object linked to the related supplier record. |
|                      | 6.                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         | Saves the payment record to the database.                                        |
|                      | 7.                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         | Displays a success confirmation message.                                         |
| Exception Conditions | <ul style="list-style-type: none"> <li>Missing Mandatory Data: If payment amount, payment date, or payment method is missing, the system displays an error message and prevents submission.</li> <li>Related Record Not Found: If the selected supplier transaction or invoice does not exist, the system prevents the payment from being saved and displays an error message.</li> </ul> |                                                                                         |                                                                                  |

|  |                                                                                                                                                                                                     |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>Invalid Amount: If the payment amount is empty, zero, or not a valid number, the system displays an error and asks the user to correct the value.</li> </ul> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.3.16.6 Class Diagram



## 6 Domain Class Diagram(s)

